

QUESTION BANK
CHAPTER 3- CURRENT ELECTRICITY

SECTION-A

1-MARK QUESTIONS

- Q1. What is meant by current and conventional current?
- Q2. Define SI unit of current.
- Q3. $1\ \Omega = 1\text{V}/1\text{A}$ From this what do you mean by 1 ohm?
- Q4. How does drift velocity of electrons change with temperature?
- Q5. What is the effect of temperature on the relaxation time of electrons in a metal?
- Q6. A wire of resistivity ρ is stretched to double its length. What will be its new resistivity?
- Q7. Which physical quantity does the voltage versus current graph for a metallic conductor depict? Give its SI unit.
- Q8. Draw the graph showing the variation of resistance of a cylindrical conductor with its radius
- Q9. Draw the graph showing the variation of resistance of a conductor with rise in temperature.
- Q10. Answer the following questions:
- (a) Alloys of metals usually have _____ resistivity than that of their constituent metals.
 - (b) Alloys usually have much _____ temperature coefficients of resistance than pure metals.
 - (c) The resistivity of the alloy manganin is _____ rapidly with increase of temperature.
 - (d) The resistivity of a typical insulator (e.g., amber) is greater than that of a metal by a factor of the order of _____
- Q11. If 2.25×10^{20} electrons pass through a wire in one minute, find the magnitude of the current flowing through the wire.
- Q12. Define EMF of a cell.
- Q13. What happens to the conductivity of electrolytes as temperature is increased?
- Q14. What are ohmic conductors? Give two examples.
- Q15. What are non ohmic devices. Give two examples.
- Q16. Draw V-I graph for GaAs.
- Q17. How is temperature coefficient of resistivity for metals is different from temperature coefficient of resistivity for semiconductor.

- Q18. Name two substances for which temperature coefficient of resistivity is positive.
- Q19. Name two materials for which temperature coefficient of resistivity is negative.
- Q20. How is temperature coefficient of resistivity different for metals and alloys.
- Q21. What happens to resistivity of metals with increase in temperature?
- Q22. What happens to resistivity of semiconductors or insulators with increase in temperature?
- Q23. Define terminal potential difference of a cell.

Multiple Choice Questions (Each 1M)

Choose the correct option(s) in the following questions.

- Q24. The resistance of a metal wire increases with increasing temperature on account of
- (a) decrease in free electron density.
 - (b) decrease in relaxation time.
 - (c) increase in mean free path.
 - (d) increase in the mass of electron.
- Q25. $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$ is the SI unit of which of the following?
- (a) Drift velocity (b) Mobility (c) Resistivity (d) Potential gradient
- Q26. Resistivity of a given conductor depends upon
- (a) temperature. (b) length of conductor. (c) area of cross-section. (d) shape of the conductor.
- Q27. The carriers of electricity in a metallic conductor are
- (a) holes (b) negative ions (c) positive ions (d) electrons
- Q28. The time rate of flow of charge through any cross section of a conductor is _____
- (a) electric potential (b) electric current (c) electric intensity (d) electric charge
- Q29. When no current is passed through a conductor:
- (a) the free electrons do not move
 - (b) the average speed of a free electron over a large period of time is not zero
 - (c) the average velocity of a free electron over a large period of time is zero
 - (d) the average of the velocities of all the free electrons at an instant is non-zero
- Q30. A steady current is flowing through a conductor of non-uniform cross-section. The charge passing through any cross-section of it per unit time is:
- (a) directly proportional to the area of cross-section

- (b) inversely proportional to the area of cross-section
- (c) proportional to square of the area of cross-section
- (d) independent of the area of cross-section

Q31. Drift velocity of electrons is due to:

- (a) motion of conduction electrons due to random collisions.
- (b) motion of conduction electrons due to electric field E .
- (c) repulsion to the conduction electrons due to inner electrons of ions.
- (d) collision of conduction electrons with each other.

Q32. Identify the set in which all the three materials are good conductors of electricity:

- (a) Cu, Ag and Au (b) Cu, Si and diamond (c) Cu, Hg and NaCl (d) Cu, Ge and Hg

Q33. The I-V characteristics shown in figure represents:

- (a) ohmic conductors (b) non-ohmic conductors (c) insulators (d) superconductors

Q34. In the equation $AB = C$, A is the current density, C is the electric field, Then B is:

- (a) resistivity (b) conductivity (c) potential difference (d) resistance

Q35. If a current of 0.5 A flows in a 60 W lamp, then the total charge passing through it in two hours will be:

- (a) 1800 C (b) 2400 C (c) 3000 C (d) 3600 C

Q36. The relaxation time in conductors:

- (a) increases with the increases of temperature (b) decreases with the increases of temperature
- (c) it does not depend on temperature (d) all of sudden changes at 400 K

Q37. A steady current of 1 A is flowing through the conductor. The number of electrons flowing through the cross-section of the conductor in 1 sec is:

- (a) 6.25×10^{15} (b) 6.25×10^{17} (c) 6.25×10^{19} (d) 6.25×10^{18}

Q38. Drift speed of electrons, when 1.5 A of current flows in a copper wire of cross-sectional area 5 mm^2 is v . If the electron density of copper is $9 \times 10^{28}/\text{m}^3$ the value of v in mm/s is close to (Take charge of electron to be $= 1.6 \times 10^{-19} \text{ C}$)

- (a) 3 (b) 0.2 (c) 2 (d) 0.02

Q39. A current of 2 mA was passed through an unknown resistor which dissipated a power of 4.4 W. Dissipated power when an ideal power supply of 11 V is connected across it is

- (a) $11 \times 10^{-4} \text{ W}$ (b) $11 \times 10^{-5} \text{ W}$ (c) $11 \times 10^5 \text{ W}$ (d) $11 \times 10^{-3} \text{ W}$

Q40. Nichrome and copper wires of same length and area of cross section are connected in series, current is passed through them, which wire gets heated first?

- (a) Nichrome (b) copper (c) both equally (d) none of them

Q41. The specific resistance of a conductor increases with

- (a) increase in temperature (b) increase in cross-sectional area
(c) decrease in length (d) decrease in cross-sectional area

Q42. Nichrome or Manganin is widely used in wire bound resistors because of their

- (a) temperature independent resistivity (b) very weak temperature dependent resistivity
(c) strong dependence of resistivity with temperature (d) mechanical strength

Q43. EMF is electromotive force, what is its SI unit:

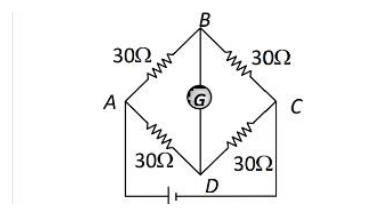
- a) Newton b) Newton sec c) Joule / coulomb d) Newton meter

Q44. Kirchhoff's laws are based upon

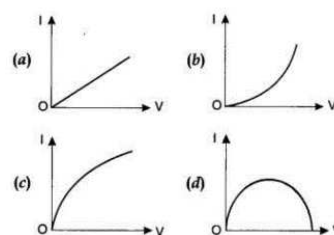
- a) Law of conservation of energy b) law of conservation of charge
c) both of these d) none of these

Q45. In the following figure current flowing through BD is

- (a) 0 (b) 0.033 A (c) 0.066 A
(d) None of these



Q46. Which of the following I-V graph represents ohmic conductors?



Q47. The specific resistance of a rod of Aluminium as compared to that of thin wire of Aluminium is:

- (a) less (b) more (c) same (d) depends upon the length and area of cross-section

Q48. A current passes through a wire of non-uniform cross section. Which of the following quantities are independent of cross section

- (a) the charge crossing (b) Drift velocity (c) current density
(d) number density of free electrons

Q49. When a battery of internal resistance 0.5 ohm is connected to a thick copper slab, a current of 12A passes through it. The emf of the cell is

- (a) 6V (b) 24V (c) 12V (d) 4V

ASSERTION AND REASONS (Each 1M)

Directions: In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as:

- (a) If both assertion and reason are true and reason is the correct explanation of assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of assertion.
(c) If assertion is true but reason is false.
(d) If both assertion and reason are false.

Q50. **Assertion:** The current density j at any point in ohmic resistor is in direction of electric field E at that point.

Reason: A point charge when released from rest in a region having only electrostatic field always moves along electric lines of force.

Q51. **Assertion:** The 200 W bulbs glows with more brightness than 100 W bulbs.

Reason: A 100 W bulb has more resistance than a 200 W bulb.

Q52. **Assertion:** Bending a wire does not affect electrical resistance.

Reason: Resistance of wire is proportional to resistivity of material.

Q53. **Assertion:** Fuse wire must have high resistance and low melting point.

Reason: Fuse is used for small current flow only.

Q54. **Assertion:** Two electric bulbs of 50 W and 100 W are given. When connected in series 50 W bulb glows more but when connected parallel 100 W bulb glows more.

Reason: In series combination, power is directly proportional to the resistance of circuit. But in parallel combination, power is inversely proportional to the resistance of the circuit.

Q55. **Assertion:** When current through a bulb decreases by 0.5%, the glow of bulb decreases by 1%.

Reason: Glow (Power) which is directly proportional to square of current.

Q56. Assertion: Two bulbs of same wattage, one having a carbon filament and the other having a metallic filament are connected in series. Metallic bulbs will glow more brightly than carbon filament bulb.

Reason: Carbon is a semiconductor.

Q57. Assertion: A current flows in a conductor only when there is an electric field within the conductor.

Reason: The drift velocity of electron in presence of electric field decreases.

Q58. Assertion: In practical application, power rating of resistance is not important.

Reason: Property of resistance remain same even at high temperature

2 MARK QUESTIONS

Q59. A potential difference of 10 V is applied across a conductor of resistance 1 k Ω . Find the number of electrons flowing through the conductor in 5 minutes.

Q60. What is drift velocity of electrons? How do you explain the flow of current in a conductor based on this?

Q61. Establish the relation between drift velocity and electric current.

Q62. What is non-Ohmic device? Give one example.

Q63. Define relaxation time of electrons in a conductor. Give its SI units. Explain, how it varies with increase in temperature of conductor?

Q64. State Ohm's law. Discuss three situations which describe the failures of Ohm's law.

Q65. A negligibly small current is passed through a wire of length 15 m and uniform cross-section $6.0 \times 10^{-7} \text{ m}^2$, and its resistance is measured to be 5.0 Ω . What is the resistivity of the material at the temperature of the experiment?

Q66. What are the factors on which the resistance of a conductor depends? Give the corresponding relation.

Q67. A potential difference of 6 V is applied across a conductor of length 0.12 m. Calculate the drift velocity of electrons, if the electron mobility is $5.6 \times 10^{-6} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ conductor.

Q68. A current of 1 A flows through a wire of length 0.24 m and area of cross-section 1.2 mm^2 , when it is connected to a battery of 3 V. Find the number density of free electrons in the wire, if the electron mobility is $4.8 \times 10^{-6} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$.

Given that charge on electron = $1.6 \times 10^{-19} \text{ C}$.

- Q69. An electron moves in a circle of radius 0.15 m with a constant speed of $3.6 \times 10^6 \text{ m s}^{-1}$. What electric current does this correspond to?
- Q70. A silver wire has a resistance of $2.1 \, \Omega$ at $27.5 \, ^\circ\text{C}$, and a resistance of $2.7 \, \Omega$ at $100 \, ^\circ\text{C}$. Determine the temperature coefficient of resistivity of silver.
- Q71. Define electric power and electric energy. Give their SI units.
- Q72. Define conductivity of a conductor. State its SI unit. Explain the variation of conductivity with temperature of a metallic conductor.
- Q73. Derive an expression for electric energy consumed in a device in terms of V, I and t , where V is the potential difference applied to it, I is the current drawn by it and t is the time for which current flows?
- Q74. Is electric current a scalar or vector quantity? Give reason.
- Q75. Why is a Wheatstone Bridge so called?
- Q76. What do you mean by sensitivity of Wheatstone Bridge? when is wheat stone Bridge most sensitive?
- Q77. What are the advantages of a Wheatstone Bridge method of measuring resistances over other methods?
- Q78. Draw a circuit of balanced Wheatstone Bridge and label it.
- Q79. Why is EMF of a cell always greater than terminal potential difference? Is there a case when terminal potential difference is greater than emf?

3 MARK QUESTIONS

- Q80. What are the factors on which internal resistance of a cell depend?
- Q81. Show the variation of resistivity of
i) copper with temperature ii) Nichrome with temperature. iii) silicon with temperature
- Q82. Two cells of different emfs and internal resistance are connected in series with one another find the expressions for equivalent EMF and equivalent internal resistance of the combination.
- Q83. A resistance R is connected across a cell of emf 8V and internal resistance r . A Voltmeter now measures the potential difference between the terminals of the cell as V . Obtain the expression for r in terms of E, V and R . Draw related circuit diagram also.

5 MARK QUESTIONS

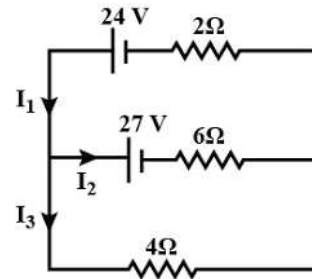
Q84. Define resistivity of a conductor and give its SI unit. Plot a graph showing the variation of resistivity with temperature for (i). Copper (ii). Nichrome (iii). Semiconductor.

How does one explain such a behavior, using the mathematical expression of the resistivity of a conductor?

Q85. Explain the variation of resistivity, of metals, semiconductors, insulators with graphs and hence define temperature coefficient of resistivity, write its SI unit.

Q86. State Kirchhoff's laws and explain them using appropriate circuit diagrams. Write about the fundamental principles on which they are based.

Q87. Applying the Kirchhoff's laws in following circuit diagram, Write all relations between I_1 , I_2 , and I_3

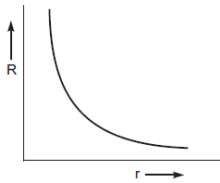


SECTION- A- ANSWER KEY

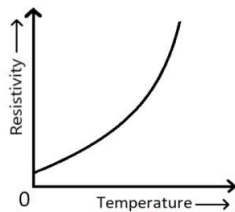
1. Rate of flow of charge is called as current. The current that flows from positive pole to negative pole of a cell in the external circuit is called conventional current.
2. One ampere is that current when one Coulomb of charge flows through the conductor in one second
- 3.If the potential difference between the ends of conductor is 1 V and if a current of 1 A flow through it, then the resistance of the conductor is 1Ω .
4. Inversely proportional to temperature
- 5.The relaxation time of electrons decreases with the rise in temperature of the metal.
- 6.The resistivity remains the same as it does not depend upon the length of the wire.
- 7.It represents resistance. It is measured in ohm.
- 8.

Resistance of a conductor of length l , and radius r is given by

$$R = \rho \frac{l}{\pi r^2}$$



9.



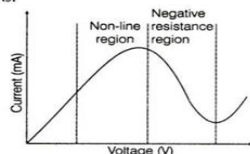
- 10.(a) greater (b) lower (c) nearly independent of (d) 10^{22}

11.0.6A

$$I = ne/t = \frac{2.25 \times 10^{20} \times 1.6 \times 10^{-19}}{60} = 0.6 \text{ A}$$

12.

Variation of current versus voltage for the material GaAs.



13. Conductivity increases due to increase in movement of ions with temperature.
14. (a) A metallic conductor for small amounts of current.
(b) Electrolytes such as CuSO_4 solution with copper electrodes.
15. Non-ohmic devices are those that do not obey Ohm's law. Vacuum tubes and thermistors are two examples.
16. The maximum potential difference which is present between two electrodes of a cell is defined as the electromotive force of a cell or EMF of a cell.
17. The temperature coefficient of resistivity is positive for metals and negative for semiconductor.
18. Gold, silver
19. silicon, germanium
20. The value of temperature co-efficient of resistance (α) is more for metals than for alloys.
21. It increases with rise in temperature.
22. Increases
23. Terminal potential difference of a cell is defined as the potential difference between the two electrodes of a cell in a closed circuit

Multiple Choice Questions –answers

- 24.(b) 25.(b) 26. (a) 27. (d) 28. (b) 29. (d) 30. (d) 31. (b) 32. (a) 33. (b) 34. (a) 35. (d) 36. (b)
37. (d)
38. As $I = neAv_d$

$$v_d = I/neA = 1.5/(9 \times 10^{28} \times 1.6 \times 10^{-19} \times 5 \times 10^{-6})$$

$$v = 0.02 \times 10^{-3} \text{ m/s} = 0.02 \text{ mm/s}$$

Answer: (d) 0.02

39. Case (1) As $I^2 R = P$

$$R = P/I^2$$

$$R = (4.4)/(2 \times 10^{-3})^2 = 1.1 \times 10^6 \Omega$$

$$\text{Case (2)} P = V^2/R = (11)^2/(1.1 \times 10^6) = 11 \times 10^{-5} \text{ W}$$

Answer: (b) $11 \times 10^{-5} \text{ W}$

40. (a) Nichrome

41. (a) Increase in temperature

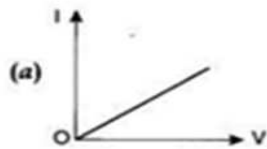
42. (b) very weak temperature dependent resistivity

43. (c) Joule / coulomb

44. (c) both of these

45. (a) 0

46. (a)



47. (c) same

48. (d) number density of free electrons

49. (a) 6V

Assuming copper slab to be ideal conductor i.e. $R = 0$

$$\Rightarrow E = I \times r$$

$$E = 12 \times 0.5$$

$$E = 6V$$

ASSERTION AND REASONS (Each 1M)

50. (C) 51. (A) 52. (A) 53. (C) 54. (A) 55. (A) 56. (D) 57. (C) 58. (D)

2 MARK QUESTIONS

59. 1.875×10^{19}

$$I = V/R = 10^{-2} \text{ A} = q/t$$

$$q = It = ne$$

$$n = 1.875 \times 10^{19}$$

60. When electrons are subjected to an electric field they do move randomly, but they slowly drift in one direction, in the direction of the electric field applied. The net velocity at which these electrons drift is known as drift velocity.

61. Let l is the length of the conductor and A uniform area of cross-section.

Therefore, the volume of the conductor $= Al$

If n is the number of free electrons per unit volume of the conductor, then the total number of free electrons in the conductor $= Aln$.

If e is the charge on each electron, then total charge on all the free electrons in the conductor
 $q = Ane$

Let a constant potential difference V is applied across the ends of the conductor with the help of a battery

The electric field set up across the conductor is given by

$$E = V/l$$

$$t = l/v_d$$

As current $I = q/t$

$$I = neAv_d$$

62. A device that does not obey Ohm's law is non ohmic device. Ex. Semiconductor diode

63. Relaxation time is the time interval between two successive collisions of electrons in a conductor, when current flows

Second Inversely proportional to temperature

64. It states that the current through any two points of the conductor is directly proportional to the potential difference applied across the conductor provided that the physical conditions remains constant.

(a) V ceases to be proportional to I .

(b) The relation between V and I depends on the sign of V . In other words, if I is the current for a certain V , then reversing the direction of V keeping its magnitude fixed, does not produce a current of the same magnitude as I in the opposite direction. This happens, for example, in a diode.

(c) The relation between V and I is not unique, i.e., there is more than one value of V for the same current I . A material exhibiting such behavior is GaAs.

65. Length $= 15 \text{ m}$

Area of cross section $6 \times 10^{-7} \text{ m}^2$

$R = 5 \text{ Ohms}$

$$\text{Resistivity } = \rho = \frac{RA}{l} = 2.0 \times 10^{-7} \Omega\text{m}$$

66. The resistance of a wire depends on its length (l), cross-sectional area (A), number of electrons per m³ (n) and the relaxation time $R = \frac{m l}{n e^2 \tau A}$

67. Ans. $2.8 \times 10^{-4} \text{ms}^{-1}$

$$E = V/l = 50 \text{ V/m}$$

$$v_d = \mu E = 2.8 \times 10^{-4} \text{ms}^{-1}$$

68. $E = V/l = 12.5 \text{ V/m}$

$$v_d = \mu E = 6 \times 10^{-5} \text{ms}^{-1}$$

$$n = \frac{I}{A e v_d} = 8.68 \times 10^{28} \text{m}^{-3}$$

69. The electron crosses a point on the circle once in every revolution. If v is number of revolutions made by the electron in 1 s, then its motion along circle corresponds to a current,

$$I = e f = e/T = \frac{e v}{2 \pi r} = 6.1 \times 10^{-13} \text{ A}$$

70. Temperature coefficient of resistivity of silver = α

$$\alpha = \frac{R_2 - R_1}{R_1 (T_2 - T_1)}$$

$$\alpha = \frac{2.7 - 2.1}{2.1 (100 - 27.5)} = 0.0039^\circ \text{C}^{-1}$$

71. **Electric energy:** The total work done by the source of emf in maintaining an electric current in a circuit for a given time is called electric energy consumed in the circuit. Its SI unit is joule.

Electric power: The rate at which work is done by a source of emf in maintaining an electric current through a circuit is called electric power of the circuit. Its SI unit is Watt.

72. Conductivity (or specific conductance) of a conductor is a measure of its ability to conduct electricity. It is also reciprocal of resistivity of the conductor. The SI unit of conductivity is Siemen per meter (S/m). The electrical conductivity of a metallic conductor will decrease with an increase in temperature.

73. Electric energy consumed = $W = Q \times V$

$$I = \frac{Q}{t}$$

$$W = I \times t \times V$$

$$\therefore W = I \times t \times V$$

$$\text{Is also} = I^2 R t \text{ and } \frac{V^2 t}{R}$$

74. Electric current is a scalar quantity the reason is that laws of ordinary algebra are used to add electric currents and laws of vector additions do not apply to the addition of electric currents.

75. It is so called because this method was first suggested by a British physicist Charles Wheatstone in 1843. It is called a bridge because galvanometer circuit forms a kind of bridge by connecting two points having the same potential.

76. Wheatstone bridge is said to be sensitive if it produces more deflection in the galvanometer for a small change of resistance in resistance arm. It is most sensitive when all four resistors P, Q, R, S are nearly of same magnitude and null point is obtained in the middle of alloy wire.

77. a) It is a null point method hence the result is free from the effect of extra resistances of the circuit.

b) As it is null point method so it is easier to detect a small change in deflection.

78. Diagram of Wheatstone bridge.

79. The EMF of a cell is greater than its terminal voltage because there is some potential drop across the cell due to its small internal resistance. During charging of a cell, terminal potential difference is greater than emf

3 MARK QUESTIONS

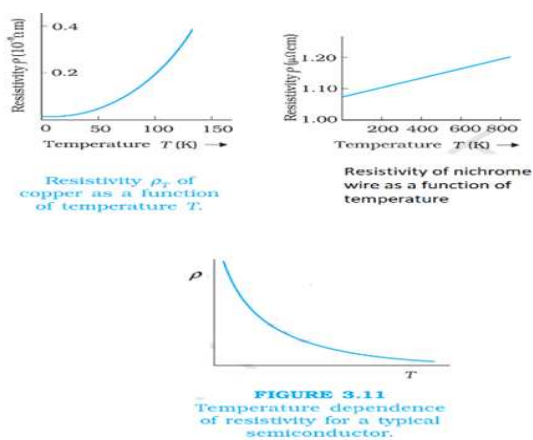
80.i. The surface area of electrodes- Larger the surface area of electrodes, less is the internal resistance.

ii. Distance between electrodes- More the distance between the electrodes, the greater is the internal resistance.

iii. Concentration of electrolyte - less ionic the electrolyte or higher the concentration of electrolyte, greater is the internal resistance.

iv. Temperature of electrolyte - higher the temperature, less is the internal resistance.

81.



82. $E_{eq} = E_1 + E_2 + E_3 \longrightarrow E_n = nE$

$r_{eq} = r_1 + r_2 + r_3 \longrightarrow r_n = nr$

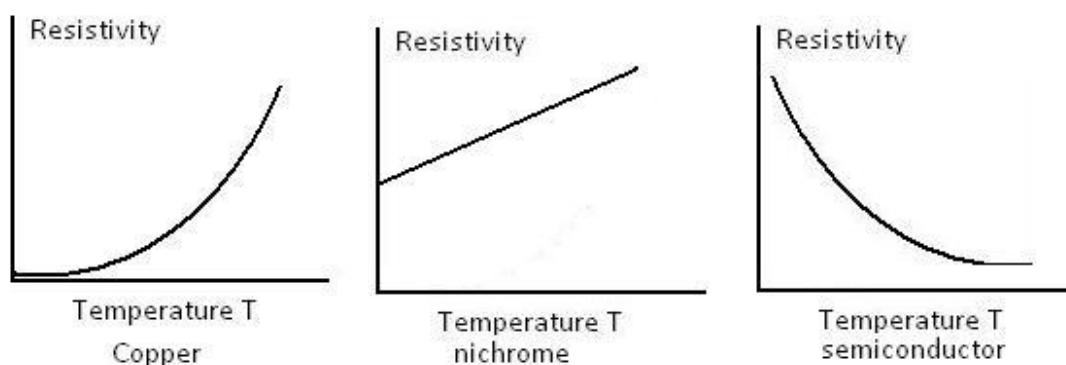
83. Obtain following all the steps $r = \frac{(E-V)R}{V}$

5 MARK QUESTIONS

84. We know that, $R = \frac{\rho l}{A}$

$l=1m, A=1m^2 \Rightarrow \rho=R$

Thus, resistivity of a material is numerically equal to the resistance of the conductor having unit length and unit cross-sectional area and its SI unit is Ohm-m



The resistivity of a material is found to be dependent on the temperature. The resistivity of a metallic conductor is given by

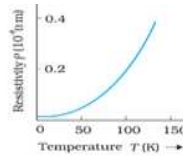
$\rho_T = \rho_0 [1 + \alpha (T - T_0)] \dots\dots\dots(i)$

Where ρ_T is the resistivity at a temperature T and ρ_0 is the same at a reference temperature T_0 , α is

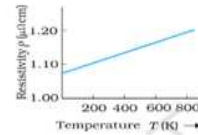
called the temperature co-efficient of resistivity. This expression shows that resistivity increases with increase in temperature of the conductor

85.Explanation using

$$\rho = \frac{1}{n\tau}$$



Resistivity ρ , of copper as a function of temperature T .



Resistivity of nichrome wire as a function of temperature

temperature coefficient of resistivity

$$\alpha = \frac{\Delta \rho}{\rho_0 \Delta T}$$

SI unit – K^{-1}



FIGURE 3.11 Temperature dependence of resistivity for a typical semiconductor.

86. Kirchhoff's laws explanations from book.

1st law is based upon law of conservation of charge and 2nd law is based upon law of conservation of energy.

87. $I_1 = I_2 + I_3$ (1) (Using KCL)

$4 I_3 + 2 I_1 - 24 = 0$ (2) (Using KVL)

$4 I_3 - 6 I_2 - 27 = 0$ (3) (Using KVL)

SECTION – B

1-Mark Questions

- Q1. Is current density scalar or vector quantity?
- Q2. Steady current is flowing in cylindrical conductor. Does electric field exist within the conductor?
- Q3. When a straight wire of resistance R bent into U-shape, does its resistance change?
- Q4. If the radius of copper wire is doubled, will its specific resistance increase, decrease or remain same?
- Q5. What is the effect of heating of conductor on the drift velocity of free electrons?
- Q6. A uniform wire resistance 50 ohms is into equal parts. These parts are now connected parallel. What the value equivalent resistance of combination?
- Q7. What happens to the power dissipation if the value of electric current passing through conductor of constant resistance doubled?
- Q8. Define the resistivity and write the SI unit.
- Q9. A wire resistivity ρ stretched double length. What will be new resistivity?
- Q10. What is the Significance of positive value of temperature coefficient of resistivity?
- Q11. What is the significance of negative value of temperature coefficient of resistivity?
- Q12. What happens to internal resistance of a cell after long use?
- Q13. Write the relation between internal resistance, EMF, and terminal potential difference of a cell
- Q14. Under what condition EMF of a cell is less than terminal potential difference.
- Q15. How does the relaxation time of electron in the conductor change when temperature of the conductor decreases.
- Q16. Resistors of high value are made up of carbon. Why?

Multiple Choice Questions (Each 1M)

- Q17. In a current carrying conductor, the ratio of the electric field and the current density at a point is called
- (a) Resistivity (b) Conductivity (c) Resistance (d) Mobility

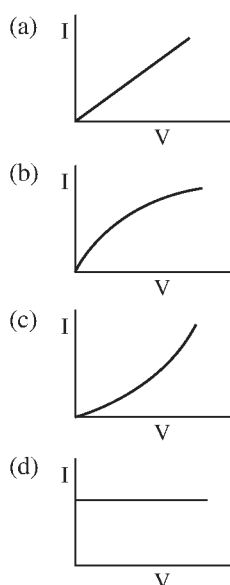
Q18. Consider a current carrying wire (current I) in the shape of a circle. Note that as the current progresses along the wire, the direction of j (current density) changes in an exact manner, while the current I remain unaffected. The agent that is essentially responsible for is _____

- (a) source of emf.
- (b) electric field produced by charges accumulated on the surface of wire.
- (c) the charges just behind the given segment of wire which push them just the right way by repulsion
- (d) the charges ahead

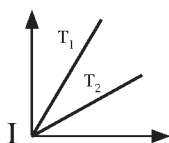
Q19. Calculate the conductance and conductivity of a wire of resistance 0.01Ω and area of cross section 10^{-4} m^2 and length 0.1 m .

- (a) 10 S and 10^5 S m^{-1}
- (b) 100 S and 10^4 S m^{-1}
- (c) 100 S and 10^6 S m^{-1}
- (d) 100 S and 10^5 S m^{-1}

Q20. The V - I characteristics of four circuit elements are shown. Which of these is ohmic?



Q21. The current-voltage (I - V) graph for a given metallic wire at two different temperatures T_1 and T_2 are shown in figure. It follows from the graph that:



V

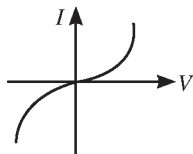
- (a) $T_1 > T_2$
- (b) $T_1 < T_2$
- (c) $T_1 = T_2$
- (d) T_1 is greater or less than T_2 depending on whether the resistance R of the wire is greater or less than the ratio V/I .

Q22. A wire has a non-uniform cross-section as shown in the figure. If a steady current is flowing through it, then the drift speed of the electrons:



- (a) is constant throughout the wire
- (b) decreases from A to B
- (c) increases from A to B
- (d) varies randomly

Q23. The I - V characteristics shown in figure represents:



- (a) ohmic conductors
- (b) non-ohmic conductors
- (c) Insulators
- (d) superconductors

Q24. A metal wire is subjected to a constant potential difference. When the temperature of the metal wire increases, the drift velocity of the electron in it:

- (a) increases, thermal velocity of electron increases
- (b) decreases, thermal velocity of electron increases
- (c) increases, thermal velocity of electron decreases
- (d) decreases, thermal velocity of electron decreases

Q25. Current of 4.8 amperes is flowing through a conductor. The number of electrons crossing any cross-section per second will be:

- (a) 3×10^{19}
- (b) 7.68×10^{21}
- (c) 7.68×10^{20}
- (d) 3×10^{20}

Q26. Ohm's law deals with the relation between:

- (a) current and potential difference
- (b) capacity and charge

- (c) capacity and potential
- (d) charge and potential difference

Q27. Ohm's law is valid when the temperature of the conductor is _____:

- (a) constant
- (b) very high
- (c) very low
- (d) varying

Q28. Ohm's law is valid for:

- (a) metallic conductors at low temperature
- (b) metallic conductors at high temperature
- (c) electrolytes when current passes through them
- (d) diode when current flows

Q29. In gallium-arsenide material, Ohm's law does not hold good because:

- (a) current remains constant for any value of voltage
- (b) resistance is infinite
- (c) negative resistance exists in the voltage-current variation
- (d) current goes to infinite at very low voltages

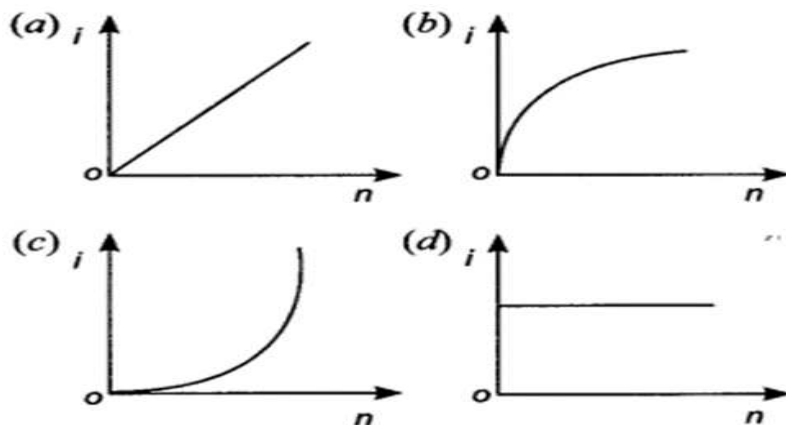
Q30. When the length and area of cross-section both are doubled, then its resistance:

- (a) will become half
- (b) will be doubled
- (c) will remain the same
- (d) will become four times

Q31. For a metallic wire, the ratio V/I (V = the applied potential difference, I = current flowing):

- (a) is independent of temperature.
- (b) increases as the temperature rises.
- (c) decreases as the temperature rises.
- (d) increases or decreases as temperature rises, depending upon the metal.

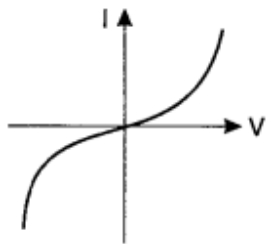
Q32. A battery consists of a variable number ' n ' of identical cells (having internal resistance ' r ' each) which are connected in series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n ?



Q33. Drift velocity v_d varies with the intensity of electric field as per the relation

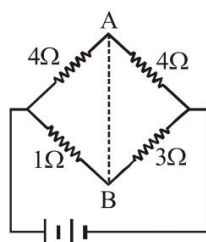
- (a) $v_d \propto E$ (b) $v_d \propto \frac{1}{E}$ (c) $v_d = \text{constant}$ (d) $v_d \propto E^2$

Q34. The I-V characteristics shown in figure represents



- (a) ohmic conductors (b) non-ohmic conductors
(c) insulators (d) superconductors

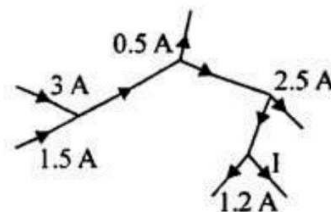
Q35. In the circuit shown, if a conducting wire is connected between points A and B, the current



in this wire will

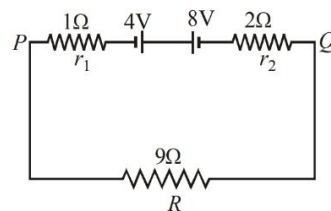
- (a) flow in the direction which will be decided by the value of V
(b) be zero
(c) flow from B to A
(d) flow from A to B

Q36. In following figure shows currents in a part of electrical circuit, then the value of I (in ampere) is given by :



- (a) 0.3 A (b) 0.5 A
(c) 1.3 A (d) None of these

Q37. Two batteries of emf 4 V and 8V with internal resistance 1 ohm and 2 ohm are connected in a circuit with a resistance of 9 W as shown in figure.



The current and potential difference between the points P and Q are

- a) $1/3$ A and 3 V (b) $1/6$ A and 4 V
(c) $1/9$ A and 9 V (d) $1/12$ A and 12 V

Q38. In a Wheatstone bridge all the four arms have equal resistance $2R$. If the resistance of the galvanometer arm A B is also R , the equivalent resistance of the combination as seen by the battery is

- (a) $2R$ (b) R (c) $2R$ (d) $4R$

Assertion-Reasoning Questions

Directions: These questions consist of two statements, each printed as Assertion and Reason. While answering these questions, you are required to choose any one of the following four responses.

- (a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
(b) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
(c) If the Assertion is correct but Reason is incorrect.
(d) If both the Assertion and Reason are incorrect.

Q39. **Assertion:** There is no current in the metals in absence of electric field.

Reason: Motion of free electrons is randomly.

Q40. **Assertion:** A wire carrying an electric current has no electric field around it.

Reason: Rate of flow of electrons in One direction is equal to rate of flow of protons in opposite direction

Q41. **Assertion:** In a simple battery circuit, the point of the lowest potential is positive terminal of the battery.

Reason: The current flows towards the point of the higher potential, as it does in such a circuit from the negative to the positive terminal

2 MARKS QUESTIONS

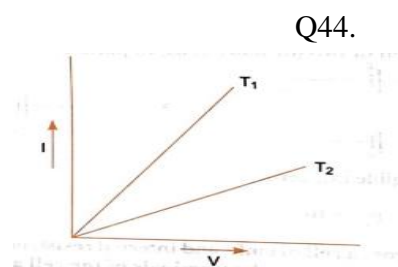
Q42. The metallic conductor is at temperature Θ_1 . The temperature of metallic conductor is increased to Θ_2 . How will the product of resistivity and conductivity change? Why?

Q43. Specific resistance copper, silver constantan are 1.18×10^{-6} , 1×10^{-6} , and 45×10^{-6} ohm cm respectively. Which is the best electrical conductor and why?

Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more? Justify your answer.

Q45. I - V graph for a metallic wire at two different temperatures,

T_1 and T_2 is as shown in the figure. Which of the two temperatures is lower and why?



Q46. What conclusion can you draw from the following observations on a resistor made of an alloy manganin?

Current A	Voltage V	Current A	Voltage V
0.2	3.94	3.0	59.2
0.4	7.87	4.0	78.8
0.6	11.8	5.0	98.6
0.8	15.7	6.0	118.5
1.0	19.7	7.0	138.2
2.0	39.4	8.0	158.0

Q47. An electric bulb rated for 500W at 100V is used in circuit having a 200V supply. Calculate the resistance R that must be put in series with the bulb, so that the bulb delivers 500W.

Q48. Why is Wheatstone Bridge method considered unsuitable for measurement of very low resistances?

Q49. Why is the Wheatstone method considered Unsuitable for measurement of very high resistances?

Q50. A Battery of emf 3 volt and internal resistance r is connected in series with 55 ohms through an ammeter of resistance 1 ohm. The ammeter reads 50 mA. Draw the circuit diagram and calculate value of r .

Q51. Derive the conditions for obtaining maximum current through external resistance connected across a series combination of n identical cells.

Q52. Derive condition for obtaining maximum current through an external resistance connected to a parallel combination of n identical cells.

Q53. Two identical cells, whether joined in series or in parallel give the same current, when connected to an external resistance of 1 ohm. Find the internal resistance of each cell.

Q54. Two wire one of copper and other of manganin have same resistance and equal length. Which wire is thicker?

Q55. When does the terminal voltage of a cell become (i) greater than its emf (ii) less than its emf?

Q56. A car battery is of 12V. Eight dry cells of 1.5 V connected in series also give 12V, but such a combination is not used to start a car. Why?

Q57. The current flowing through a conductor is 2mA at 50V and 3mA at 60V. Is it an ohmic or non-ohmic conductor? Give reason.

3MARKS QUESTIONS

Q58. On what factors resistivity of the material depend? Write the corresponding equation. Why copper wires are used as connecting wires?

Q59. A heating element using nichrome connected to a 230 V supply draws an initial current of 3.2 A which settles after a few seconds to a steady value of 2.8 A. What is the steady temperature of the heating element if the room temperature is 27.0°C ? Temperature coefficient of resistance of nichrome averaged over the temperature range involved is $1.70 \times 10^{-4}^{\circ}\text{C}^{-1}$?

Q60. Answer the following questions:

(a) A steady current flows in a metallic conductor of the non-uniform cross-section. Which of these quantities is constant along the conductor: current, current density, electric field, drift speed?

(b) Is Ohm's law universally applicable for all conducting elements? If not, give examples of

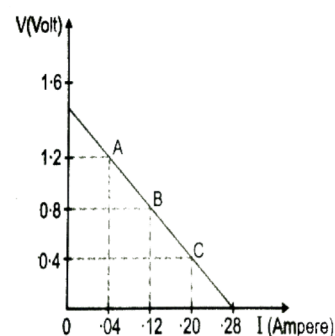
elements that do not obey Ohm's law.

(c) A low voltage supply from which one needs high currents must have very low internal resistance. Why?

Q61 . Two cells of different emfs and internal resistance are connected in parallel with one another find the expressions for equivalent EMF and equivalent internal resistance of the combination.

Q62. A battery of emf 12.0 V and internal resistance 0.5Ω is to be charged by a battery charger which supplies 110 V dc. How much resistance must be connected in series with the battery to limit the charging current to 5.0 A? What will be the potential difference across the terminals of the battery during charging?

Q63. Potential differences across the terminals of a cell were measured (in volt) against different currents (in ampere) flowing through the cell. A graph was drawn which was a straight-line ABC as shown in figure

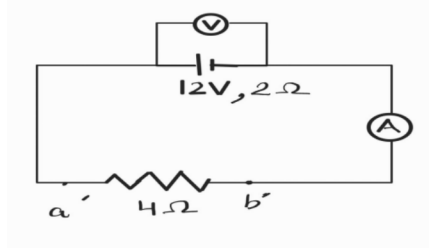


Determine from graph (i) emf of the cell (ii) maximum current obtained from the cell and (iii) internal resistance of the cell.

Q64. Four identical cells each of emf 2V, are joined in parallel providing supply of current to external circuit consisting of two 15Ω resistors joined in parallel. The terminal voltage of the cells as read by an ideal voltmeter is 1.6V. Calculate the internal resistance of each cell.

Q65. Give three differences between e.m.f. and terminal potential difference of a cell.

Q66. In the figure, an ammeter A, and a resistor of resistance $R = 4\Omega$ have been connected to the terminals of the source to form a complete circuit. The emf of the source is 12V having an internal resistance of 2Ω . Calculate voltmeter and ammeter reading.



5 MARKS QUESTIONS

Q67. (a) On the basis of electron drift, derive an expression for resistivity of a conductor in terms of number density of free electrons and relaxation time. On what factors does resistance and resistivity of a conductor depend?

(b) Why alloys like constantan and manganin are used for making standard resistors?

Q68.(a) The electron drift arises due to the force experienced by electrons in the electric field inside the conductor. But force should cause acceleration. Why then do the electrons acquire a steady average drift speed?

(b) If the electron drift speed is so small, and the electron's charge is small, how can we still obtain large amounts of current in a conductor?

(c) When electrons drift in a metal from lower to higher potential, does it mean that all the 'free' electrons of the metal are moving in the same direction?

(d) Are the paths of electrons straight lines between successive collisions (with the positive ions of the metal) in the (i) absence of electric field, (ii) presence of electric field?

Q69. Define current density. Give its SI unit. Whether it is vector or scalar?

How does it vary when (i) potential difference across wire increases (ii) length of wire increases (iii) temperature of wire increases (iv) Area of cross-section of wire increases justify your answer.

Q70. State Kirchhoff's rules for electrical networks. Use them to explain the principle of Wheatstone bridge for determining an unknown resistance. How is it realized in actual practice in the laboratory? Write the formula used.

Q71. Deduce the condition for balance in a Wheatstone bridge. Write any two important precautions you would observe while performing the experiment. When is Wheatstone bridge most sensitive?

SECTION-B- Answers

1. Current density is vector quantity.
2. Yes, electric field exists within the conductor because it is the electric field which imparts acceleration to electrons for the flow of current.
3. No, the resistance remains same, because length and cross-sectional area of the wire remain unchanged.
4. The specific resistance of wire depends only on the material (at given temperature). Therefore by changing the radius, the specific resistance of copper remains unchanged.
5. $V_d = eE \frac{\tau}{m}$

By heating conductor, the collisions electrons occur more frequently; relaxation time decreases and hence drift velocity decreases.
6. 2 ohms
7. Power $P = I^2 R t \propto I^2$

Clearly current doubled, power dissipated becomes times.
8. The resistivity of material of conductor defined the resistance offered by conductor length 1m and area cross-section 1 m². Its unit ohm metre
9. New resistivity will be (unchanged) because resistivity independent of dimensions
10. Resistivity increases with increase in temperature.
11. Resistivity decreases with increase in temperature.
12. As the battery or cell is in continuous use, consuming electrolyte and the chemical reaction keeps occurring in the cell, due to which the concentration of ions in the cell decreases and this resists the flow of charge through it. Hence, the internal resistance increases with time.
13. $E = V + Ir$.
14. During charging.
15. When temperature of the conductor decreases, ionic vibration in the conductor decreases so relaxation time increases.
16. High resistivity and low temperature Coefficient of resistance.

Multiple Choice Questions (Each 1M) -solution

17.(a)

18.(b)

19. (d) 100 S and 10^5 S m^{-1}

$$R = \frac{\rho l}{A}$$

$$G = 1/R = 100 \text{ S}$$

$$\text{Conductivity} = 1/\text{Resistivity} = 10^5 \text{ S m}^{-1}$$

20.(a)

21.(b)

22. (b)

23. (b)

24.(b)

25. (a)

26. (a)

27. (a)

28. (a)

29. (c)

30. (c)

31. (b)

32. (d)

Here, $i = nE/n$, $r = E/r$ Because I is totally independent of n , hence it will remain constant.

33. (a) $v_d \propto E$

34. (b)

35. (c) flow from B to A

36. (a) 0.3 A

37. (a)

$$I = \frac{8-4}{1+2+9} = \frac{4}{12} = 1/3 \text{ A}$$

$$V_P - V_Q = \left(4 - \frac{1}{3} \times 3\right) = 3 \text{ V}$$

38. (a) 2 R

39

40.

41.

2 MARKS QUESTIONS -ANSWERS

42. Product of resistivity and conductivity is constant = 1

Product independent temperature.

43. Smaller the resistivity of substance, larger is conductivity. The resistivity of silver is least so silver is the best conductor

44. Nichrome wire gets heated up more. Heat dissipated in a wire is given by

$$H = I^2 R t = I^2 \frac{\rho l}{A} t$$

$$H \propto \rho$$

ρ of nichrome > ρ of copper

H of nichrome > H of copper

45. Answer

$$R_1 = V_1/I \quad \text{and} \quad R_2 = V_2/I$$

$$V_2 > V_1 \text{ So } R_2 > R_1 \quad \text{and} \quad T_1 < T_2$$

46. Ohm's law is valid to high accuracy. This means that the resistivity of the alloy manganin is nearly independent of temperature.

47. Resistance of bulb = $V^2/P = 20 \Omega$, $I = 5A$, for the same power dissipation, current should be 5A when the bulb is connected to a 200V supply. The safe resistance $R' = V/I = 40\Omega$. Therefore, 20Ω resistors should be connected in series.

48. For measuring lower resistance all other resistances should have low value so that bridge is sensitive, this requires galvanometer of very low resistance, also the end resistance and resistance of connecting wires becomes comparable to the resistance measured hence error may be introduced.

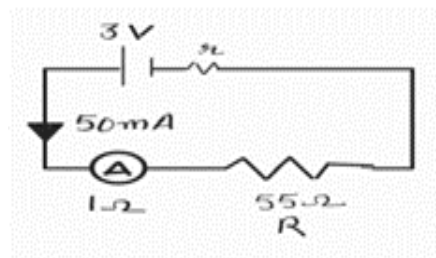
49. For measuring higher resistance all other resistances forming the wheat stone bridge should be high but this reduces the current through the galvanometer and hence it becomes insensitive.

50. $\text{Emf} = 3 \text{ V}$

$$R_{\text{net}} = 55 + 1 + r = 56 + r$$

$$I = 50 \times 10^{-3} \text{ A}$$

$$I = E/R_{\text{net}}$$



$$R = 4 \text{ ohm}$$

$$51. I = \frac{nE}{R + nr}$$

If $R \gg nr$

$$I = \frac{nE}{R} = n \times \text{current from a single cell}$$

When external resistance is much higher

than the total internal resistance then the cells should be connected in series to get maximum current.

52.

$$I = \frac{\mathcal{E}}{R + \frac{r}{n}}$$

$$I = \frac{n\mathcal{E}}{nR + r}$$

$$\text{IF } \frac{r}{n} \gg R$$

$$r \gg nR$$

$$I = \frac{nE}{r} = n \times \text{current from a single cell}$$

When external resistance is much smaller than the total internal resistance then the cells should be connected in parallel to get maximum current.

$$53. E_{\text{Series}} = 2E$$

$$E_{\text{Parallel}} = E$$

$$I = \frac{2E}{2r+1}$$

$$I = \frac{E}{\frac{r}{2}+1}$$

$$\frac{2E}{2r+1} = \frac{E}{\frac{r}{2}+1}$$

$$\frac{2}{2r+1} = \frac{1}{\frac{r}{2}+1}$$

$$r = 1 \text{ ohm}$$

54. where ρ = specific resistance of wire

$$\text{for copper wire } R = \rho_c L / A_c,$$

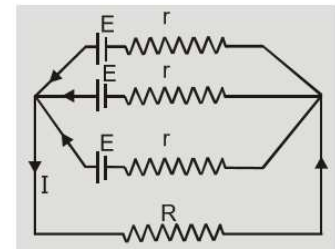
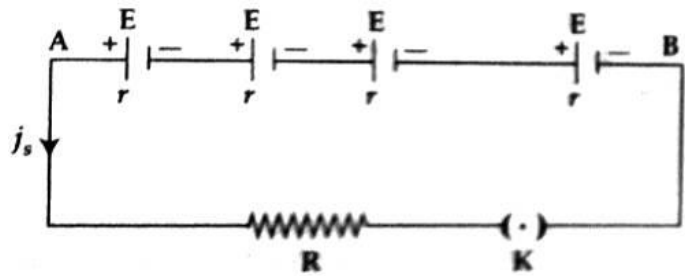
$$\text{for manganin wire } R = \rho_m L / A_m$$

$$\text{or } \rho_c / A_c = \rho_m / A_m$$

$$\text{or } A_c / A_m = \rho_c / \rho_m$$

we know $\rho_m > \rho_c$ (as manganin is an alloy)

therefore $A_c < A_m$ Hence manganin wire is thicker



55. (i) When the cell is being charged terminal potential difference (V) becomes greater than emf (E), $V = E + Ir$

(ii) When the cell is discharged, then $V < E$, $V = E - Ir$

56. Dry cell used in series will have high resistance ($= 10\Omega$) and hence provide low current, while a car battery has low internal resistance (0.1Ω) and hence gives high current for the same emf, needed to start the car.

57. The resistance of first case, $R_1 = V_1 / I_1 = 50 / 2 \times 10^{-3} = 25,000 \Omega$ the resistance of second case, $R_2 = V_2 / I_2 = 60 / 3 \times 10^{-3} = 20,000 \Omega$

As resistance changes with current, so the given conductor is non-ohmic.

3 MARKS QUESTIONS

58. The resistivity of a conductor depends on two factors i.e., nature of the material of the conductor and temperature.

$$\rho = \frac{m}{ne^2\tau}$$

Copper wires are used as connecting wires because the electrical resistivity of copper is low. It prevents any wastage of electric current.

59. The supply voltage is $V = 230 \text{ V}$

The initial current drawn is $I_1 = 3.2 \text{ A}$

Consider the initial resistance to be R_1 , which can be found by the following relation:

$$R = V/I = 71.87 \Omega$$

Value of current at steady state, $I_2 = 2.8 \text{ A}$

Value of resistance at steady state = R_2

R_2 can be calculated by the following equation :

$$R_2 = 82.14 \Omega$$

The temperature coefficient of nichrome averaged over the temperature range involved is

$$1.70 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}$$

Value of initial temperature of nichrome, $T_1 = 27.0 \text{ } ^\circ\text{C}$

Value of steady state temperature reached by nichrome = T_2

This temperature T_2 can be obtained by the following formula

$$\alpha = \frac{R_2 - R_1}{R_1(T_2 - T_1)}$$

$$T_2 - 27 = \frac{82.14 - 71.87}{71.87 \times (1.7 \times 10^{-4})}$$

$$T_2 - 27 = 840.5$$

$$T_2 = 840.5 + 27 = 867.5^\circ \text{C}$$

60. (a) Current is given to be steady. Therefore, it is a constant. The current density, electric field, drift speed depends on the area of cross-section inversely.

(b) No, examples of non-ohmic elements are vacuum diode, semiconductor diode etc.

(c) Because the maximum current drawn from a source = $\mathcal{E}/r$.

$$61. E_{eq} = (\mathcal{E}_2 r_1 + \mathcal{E}_1 r_2) / (r_1 + r_2)$$

$$(1/R_{eq}) = (1/R_1) + (1/R_2) + (1/R_3) + \dots + (1/R_n)$$

$$62. E = 110 - 12 = 98 \text{ V}$$

$$I = \frac{98}{R + 0.5}, R = 19.1 \text{ ohm}, V = E + ir = 12 + 5 \times 0.5 = 14.5 \text{ Volt}$$

$$63. (i) 1.4 \text{ V} \quad (ii) I_{\max} = 0.28 \text{ A} \quad (iii) r = E/I_{\max} = 1.4/0.28 = 5 \text{ ohm}$$

$$64. E = 2 \text{ V}$$

$$r_{eq} = r/4$$

$$R_{ex} = R/2 = 7.5$$

$$V = 1.6 \text{ V}$$

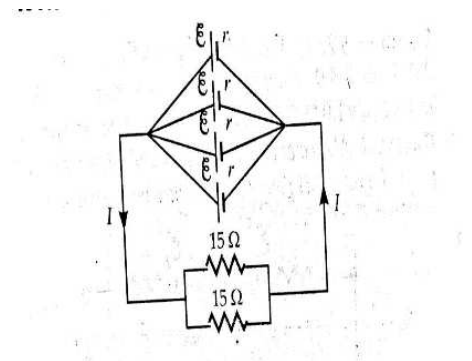
$$E = V + I r_{eq}$$

$$E - V = I r_{eq} \quad I = V / R_{ex}$$

$$r_{eq} = \frac{(E - V) R_{ex}}{V} = \frac{(2 - 1.6) 5}{1.6} = 15/8 \text{ ohm}$$

$$\frac{r}{4} = \frac{15}{8}$$

$$r = 7.5 \text{ ohm}$$



65.

E.M.F. OF A CELL	TERMINAL VOLTAGE OF A CELL
1.It is measured by the amount of work done in moving a unit positive charge in the complete circuit inside and outside the cell.	1.It is measured by the amount of work done in moving a unit positive charge in the circuit outside the cell.
2.It is the characteristic of the cell i.e.; it does not depend on the amount of current drawn from the cell	2.It depends on the amount of current drawn from the cell. More the current is drawn from the cell, less is the terminal voltage
3.It is equal to the terminal voltage when cell is not in use while greater than the terminal voltage when cell is in use.	3.It is equal to the emf of cell when cell is not in use, while less than the emf when cell is in use.

66. $E = 12 \text{ V}$ $R_{\text{net}} = 2 + 4 = 6 \text{ ohm}$

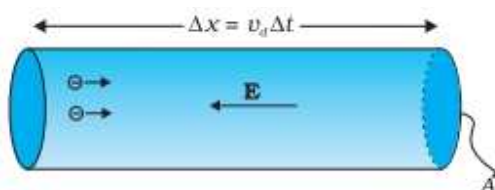
$I = 12 / 6 = 2 \text{ A}$

VOLTMETER READING = $I R = 2 \times 4 = 8 \text{ V}$,

Ammeter reading = 2 A

5 MARKS QUESTIONS

67.



Relaxation time of free electrons drifting in a conductor is the average time elapsed between two successive collisions.

Consider a conductor of length l and cross-sectional area A . When a potential difference V is applied across its ends, the current produced is I .

If n is the number of electrons per unit volume in the conductor and v_d the drift velocity electrons, then the relation between current and drift velocity is

$$I = -neAv_d$$

Where $-e$ is the charge on electron

Electric field produced at each point of wire, $E = \frac{V}{l}$

If τ is relaxation time and E is electric field strength, then drift velocity

$$v_d = -\frac{e\tau E}{m}$$

$$I = -neA\left(-\frac{e\tau E}{m}\right) \text{ or } I = \frac{ne^2\tau}{m} AE$$

$$\text{As } E = \frac{V}{l}$$

$$I = \frac{ne^2}{m} \tau A \frac{V}{l} \text{ or } \frac{V}{l} = \frac{m}{ne^2\tau A} \frac{I}{A}$$

$$\text{Current density } J = \frac{I}{A} = \frac{ne^2\tau}{ml} V$$

This is relation between current density J and applied potential difference V .

Under give physical conditions (temperature, pressure) for a given conductor

$$\frac{m}{ne^2\tau A} \frac{l}{A} = \text{Constant} \quad \therefore \text{This constant is called the resistance of the conductor (i.e. } R)$$

$$\text{i.e. } R = \frac{m}{ne^2\tau A} \frac{l}{A} \text{-----1}$$

$$\frac{V}{l} = R$$

This is Ohm's law from equation it is clear that the resistance of a wire depends on its length (l), cross-sectional area (A), number of electrons per m^3 (n) and the relaxation time

Expression for resistivity

$$\text{As } R = \frac{\rho l}{A} \text{-----2}$$

Comparing 1 and 2, we get

$$\text{Resistivity of a conductor } \rho = \frac{m}{ne^2\tau}$$

clearly, resistivity of a conductor is inversely proportional to number density of free electrons and relaxation time.

Resistivity of the material of a conductor depends upon the relaxation time - i.e. temperature and the number density of free electrons. This is because constantan and manganin show very weak dependence of resistivity on temperature.

68. (a) Each 'free' electron does accelerate, increasing its drift speed until it collides with a positive ion of the metal. It loses its drift speed after collision but starts to accelerate and increases its drift speed again only to suffer a collision again and so on. On the average, therefore, electrons acquire only a drift speed.

(b) Simple, because the electron number density is enormous, $\sim 10^{29} \text{ m}^{-3}$.

(c) By no means. The drift velocity is superposed over the large random velocities of electrons.

(d) In the absence of electric field, the paths are straight lines; in the presence of electric field, the paths are, in general, curved

69. Definition, SI unit is A / m^2

Current density is a vector quantity. Its direction is same as that of motion of positive charge.

$$J = \frac{I}{A} = \frac{neAv}{A} = \frac{neeE\tau}{m} = nee \frac{V\tau}{ml} = ne^2 \frac{V\tau}{ml}$$

(a) With increase in potential gradient, J increase.

(b) With increase in temperature, τ decreases, so J decreases.

(c) With increase in length J decreases.

(d) With increase in area, J remains unchanged as J is independent of A.

70. DERIVATION, CIRCUIT DIAGRAM

71. The current should not flow in the Set up for a long time, otherwise the wire will become hot and its resistance will be changed.

The null point should be between 45cm and 55cm.

It is most sensitive When all four registers P,Q,R,S are nearly of same magnitude and null point is obtained in the middle of alloy wire.

SECTION-C

Multiple Choice Questions (Each 1M)

1. Consider a current carrying wire (current I) in the shape of a circle. Note that as the current progresses along the wire, the direction of j (current density) changes in an exact manner, while the current I remain unaffected. The agent that is essentially responsible for is:

- (a) source of emf
- (b) electric field produced by charges accumulated on the surface of wire.
- (c) the charges just behind a given segment of wire which push them just the right way by repulsion.
- (d) the charges ahead

2. A metal rod of length 10 cm and a rectangular cross section of $1\text{ cm} \times 1/2\text{ cm}$ is connected to a battery across opposite faces. The resistance will be:

- (a) maximum when the battery is connected across $1\text{ cm} \times 1/2\text{ cm}$ faces
- (b) maximum when the battery is connected across $10\text{ cm} \times 1\text{ cm}$ faces
- (c) maximum when the battery is connected across $10\text{ cm} \times 1/2\text{ cm}$ faces
- (d) same irrespective of the three faces

3. Which of the following characteristics of electrons determines the current in a conductor?

- (a) Drift velocity alone
- (b) Thermal velocity alone
- (c) Both drift velocity and thermal velocity

4. A constant voltage is applied between two ends of a metallic wire. If the length is halved and the radius of the wire is doubled, the rate of heat developed in the wire will be

- (a) Increased 8 times
- (b) Unchanged
- (c) Doubled
- (d) Halved

5. A heating element has a resistance of $100\ \Omega$ at room temperature. When it is connected to a supply of 220 V, a steady current of 2 A passes in it and the temperature is 500°C more than room temperature. What is the temperature coefficient of resistance of the heating element?

- (a) $1 \times 10^{-4}\ ^\circ\text{C}^{-1}$
- (b) $2 \times 10^{-4}\ ^\circ\text{C}^{-1}$
- (c) $0.5 \times 10^{-4}\ ^\circ\text{C}^{-1}$
- (d) $5 \times 10^{-4}\ ^\circ\text{C}^{-1}$

6. A uniform wire of length l and radius r has a resistance of $100\ \Omega$. It is recast into a wire of radius $r/2$. The resistance of new wire will be

- (a) $400\ \Omega$
- (b) $100\ \Omega$
- (c) $200\ \Omega$
- (d) $1600\ \Omega$

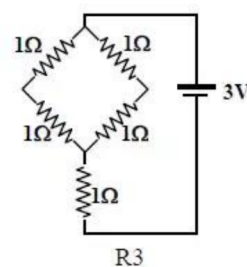
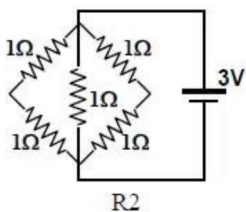
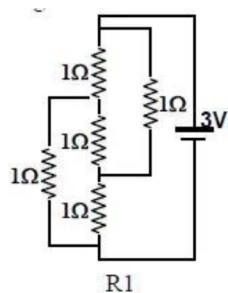
7. In a large building, there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 heater of 1 kW. The voltage of the electric mains is 220 V. The minimum capacity of the main fuse of the building will be

- (a) 14 A (b) 8 A (c) 10 A (d) 12 A

8. If a wire is stretched to make it 0.1% longer, its resistance will

- (a) increase by 0.05% (b) increase by 0.2%
(c) decrease by 0.2% (d) decrease by 0.05%

9. Figure shows three resistor configurations R1, R2 and R3 connected to 3 V battery.

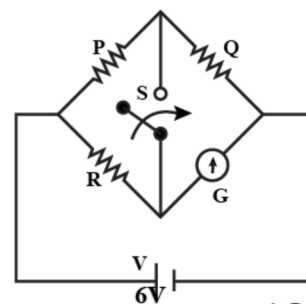


If the power dissipated by the configuration R1, R2 and R3 is P_1 , P_2 and P_3 , respectively, then

- (a) $P_1 > P_2 > P_3$ (b) $P_1 > P_3 > P_2$ (c) $P_2 > P_1 > P_3$ (d) $P_3 > P_2 > P_1$

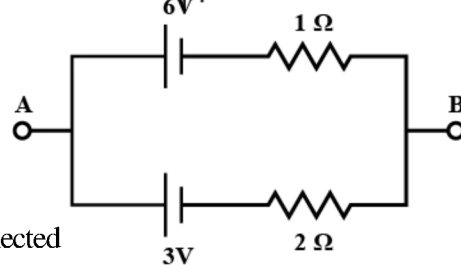
10. In the circuit shown, P is not equal to R, the reading of the galvanometer is same with switch S open or closed. Then,

- (a) $I_R = I_G$ (b) $I_P = I_G$ (c) $I_Q = I_G$ (d) $I_Q = I_R$



11. Two batteries of different emfs and different internal resistances are connected as shown. The voltage across AB in volts is

- (a) 4 (b) 5 (c) 6 (d) 7



12. Ten identical cells each of potential E and internal resistance r are connected in series to form a closed circuit. An ideal voltmeter connected across 3 cells will read

- (a) 10E (b) 13E (c) 3E (d) 5E

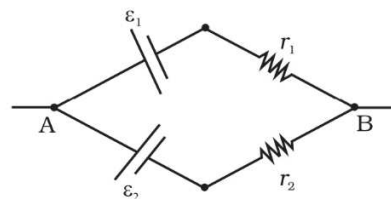
13. The resistance of a wire at a temperature t and 0 degree are related as

- (a) $R = R_0(1 + BT)$ (b) $R = R_0(1 - BT)$ (c) $R = R_0(1 + B^2T)$ (d) $R = R_0(1 - B^2T)$

14. Temperature dependence of resistivity $\rho(T)$ of semiconductors, insulators and metals is significantly based on the following factors:

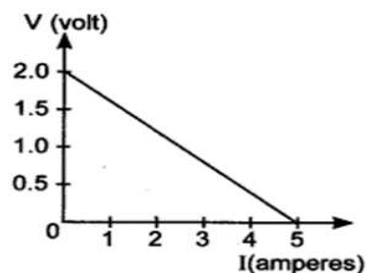
- (a) number of charge carriers can change with temperature T .
- (b) time interval between two successive collisions can depend on T .
- (c) length of material can be a function of T .
- (d) mass of carriers is a function of T .

15. Two batteries of emf ϵ_1 and ϵ_2 ($\epsilon_2 > \epsilon_1$) and internal resistances r_1 and r_2 respectively are connected in parallel as shown



- (a) The equivalent emf ϵ_{eq} of the two cells is between ϵ_1 and ϵ_2 , i.e., $\epsilon_1 < \epsilon_{eq} < \epsilon_2$
- (b) The equivalent emf ϵ_{eq} is smaller than ϵ_1 .
- (c) The ϵ_{eq} is given by $\epsilon_{eq} = \epsilon_1 + \epsilon_2$ always.
- (d) ϵ_{eq} is independent of internal resistances r_1 and r_2 .

16. For a cell, the graph between the potential difference (V) across the terminals of the cell and the current (I) drawn from the cell is shown in the figure

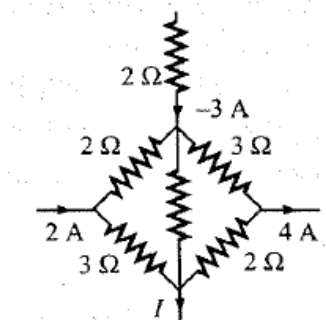


The e.m.f and the internal resistance of the cell are

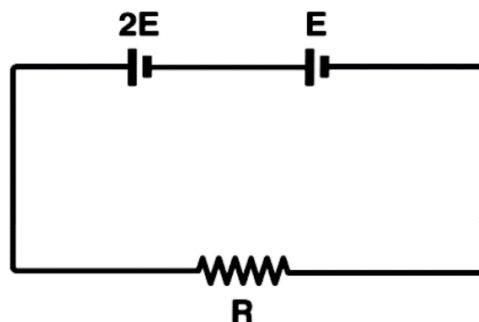
- (a) 2V, 0.5 Ω
- (b) 2V, 0.4 Ω
- (c) > 2V, 0.5 Ω
- (d) > 2V, 0.4 Ω

17. What is the current I for the circuit shown in the figure?

- (a) 3A
- (b) 1A
- (c) -5A
- (d) 5A



18. Two cells of emf $2E$ and E with internal resistance r_1 and r_2 respectively are connected in series to an external resistor R (see figure). The value of R , at which the potential difference across the terminals of the first cell becomes zero is



- (a) $r_1 - r_2$
- (b) $r_1 + r_2$
- (c) $(r_1/2) + r_2$
- (d) $(r_1/2) - r$

V.S.A(1 MARK)

Q19. How the internal resistance of a cell changes when area of the anode is decreased?

Q20. Why are alloys used for making standard resistances?

Q21. Write the dimensional formula of emf of a cell.

Q22. Alloys of metal have greater resistivity than that of their constituent metals, why?

DIRECTIONS:(a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(b) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(c) If the Assertion is correct but Reason is incorrect.

(d) If both the Assertion and Reason are incorrect.

Q23.**Assertion:** The current density J at a point in ohmic resistor is in direction of electric field E at that point

Reason: A point charge when released from rest in a region having only electrostatic field always moves along electric lines of force

DIRECTIONS: (a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(b) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(c) If the Assertion is correct but Reason is incorrect.

(d) If the Assertion is correct but Reason is incorrect.

Q24. **Assertion:** The temperature dependence of resistance is usually given as $R=R_0(1+\alpha\Delta t)$.

The resistance of wire changes from 100 ohm to 150 ohm when its temperature is increased from 27 degree Celsius to 227 degrees Celsius. This implies that $\alpha = 2.5 \times 10^{-3}$ /degree Celsius.

Reason: $R=R_0(1+ \alpha\Delta t)$ is valid only when the change in temperature is small and $\delta R \ll R_0$

DIRECTIONS: (a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(b) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(c) If the Assertion is correct but Reason is incorrect.

(d) If the Assertion is correct but Reason is incorrect.

Q 25. Assertion: EMF is potential difference between two terminals of the cells when no current is drawn from it.

Reason: It is an effect of terminal potential difference.

DIRECTIONS: (a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(b) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(c) If the Assertion is correct but Reason is incorrect.

(d) If the Assertion is correct but Reason is incorrect.

Q26. Assertion: The number density of free electrons in metals is very high.

Reason: Number density of metals increases with increase in temperature hence it is the cause of increase in conductivity of metals with temperature.

DIRECTIONS: (a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(b) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(c) If the Assertion is correct but Reason is incorrect.

(d) If the Assertion is correct but Reason is incorrect.

Q27. Assertion: Internal resistance of a cell decreases when immersed area of electrodes in the electrolyte increases

Reason: Internal resistance of a cell is its characteristic property.

DIRECTIONS: (a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(b) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(c) If the Assertion is correct but Reason is incorrect.

(d) If the Assertion is correct but Reason is incorrect.

Q28. Assertion: Which of Junction rule reveals that at a junction sum of current is zero.

Reason: Kirchhoff's Junction rule is based upon law of conservation of charge.

2 MARK QUESTIONS

Q29. A current of 2.4 A flows through a wire of cross-sectional area 1.5 mm^2 . Find the current density in the wire. If the wire contains 8×10^{28} free electrons calculate the drift velocity of electrons per cubic meter, calculate the drift velocity of electrons.

Q30. An aluminum wire of diameter 0.24 cm is connected in series to a copper wire of diameter 0.16 cm. The wires carry an electric current of 10 A. Find (a) current density of free electrons in the aluminum wire and (b) drift velocity of electrons in the copper wire. Given that number of densities of free electrons in copper $= 8.4 \times 10^{28} \text{ m}^{-3}$.

Q31. Given that resistivity of copper is $1.68 \times 10^{-8} \Omega \text{ m}$.

Calculate the amount of copper required to draw a wire 10 km long having resistance 10 ohm. The density of copper is $8.9 \times 10^3 \text{ kg m}^{-3}$.

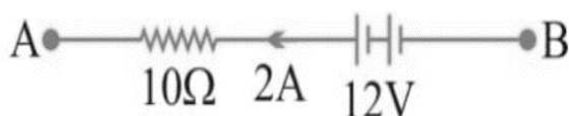
Q32. Which lamp has greater resistance (i) 60W and (ii) 100W when connected to the same supply? Why?

Q33. Nichrome and Cu wires of the same length and same diameter are connected in series in an electric circuit. In which wire will the heat be produced at a higher rate? Give reason.

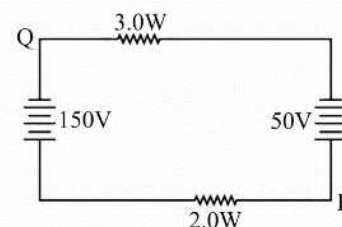
Q34. The resistance of the platinum wire of a platinum resistance thermometer at the ice point is 5Ω and at steam point is 5.23Ω . When the thermometer is inserted in a hot bath, the resistance of the platinum wire is 5.795Ω . Calculate the temperature of the bath.

Q35. A rheostat has 100 turns of a wire of radius 0.4 mm having resistivity $4.2 \times 10^{-7} \Omega \text{ m}$. The diameter of each turn is 3 cm. What is the maximum value of resistance it can introduce?

Q36. In the figure, what is the potential difference between A and B?

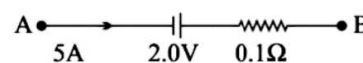


Q37. In the figure, if the potential at point P is 100V, what is the potential at point Q?



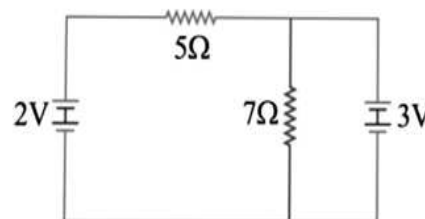
Q38. A battery of emf 2.0 V and internal resistance 0.1Ω is being charged with a current of 5.0 A.

What is the potential difference between the terminals of the battery?

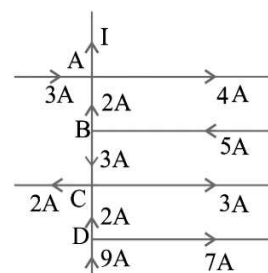


Q39. Two resistance 5Ω and 7Ω are joined as shown to two batteries of emf 2V and 3V. If the 3V battery is short circuited.

What will be the current through 5Ω ?



Q40. Find the value of I in the given circuit:



Q41. Is the momentum conserved when charge crosses a junction in an electric circuit? Why or why not?

3 MARK QUESTIONS

Q42. (a) Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area $1.0 \times 10^{-7} \text{ m}^2$ carrying a current of 1.5 A. Assume that each copper atom contributes roughly one conduction electron. The density of copper is $9.0 \times 10^3 \text{ kg/m}^3$, and its atomic mass is 63.5 u.

(b) Compare the drift speed obtained above with, (i) thermal speeds of copper atoms at ordinary temperatures, (ii) speed of propagation of electric field along the conductor which causes the drift motion.

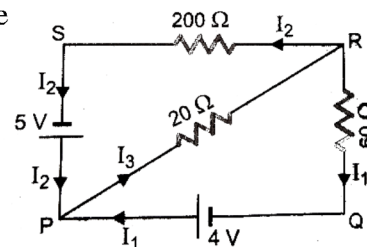
Q43. Two wires of equal length, one of aluminium and the other of copper have the same resistance. Which of the two wires is lighter? Hence explain why aluminium wires are preferred for overhead power cables.

($\rho_{\text{Al}} = 2.63 \times 10^{-8} \Omega \text{ m}$, $\rho_{\text{Cu}} = 1.72 \times 10^{-8} \Omega \text{ m}$, Relative density of Al = 2.7, of Cu = 8.9.)

Q44. An electric toaster uses nichrome for its heating element. When a negligibly small current passes through it, its resistance at room temperature (27.0°C) is found to be 75.3Ω . When the toaster is connected to a 230 V supply, the current settles, after a few seconds, to a steady value of 2.68 A. What is the steady temperature of the nichrome element? The temperature coefficient of resistance of nichrome averaged over the temperature range involved, is $1.70 \times 10^{-4} ^\circ\text{C}^{-1}$.

Q45. The amount of charge passing through the cross-section of a wire in time t is given by $q=at^2+bt+c$ (a) What are the dimensional formulae of constants a , b and c ? (b) If the values of constants a , b , c are 3, 5 and 2 in SI units, find the value of current at $t=3s$.

Q46. Apply Kirchhoff's rules to the length PRSP and PRQP to find the current I_1 , I_2 and I_3 in the circuit.

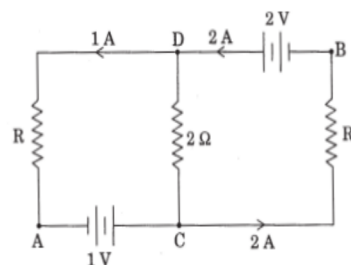


Q47. A cell of e.m.f. ' E ' and internal resistance ' r ' is connected across a variable resistor ' R '. Plot a graph showing the variation of terminal potential ' V ' with resistance ' R '. Predict from the graph the condition under which ' V ' becomes equal to ' E '. What is significance of this graph?

Q48. Four cells each of internal resistance 0.8Ω and emf $1.4V$, are connected (i) in series (ii) in parallel. The terminals of the battery are joined to the lamp of resistance 10Ω . Find the current through the lamp and each cell in both the cases.

Q49. When resistance of 2Ω is connected across the terminals of a battery, the current is $0.5A$. When the resistance across the terminal is 5Ω , the current is $0.25A$. (i) Determine the emf of the battery (ii) What will be current drawn from the cell when it is short circuited.

Q50. In the given circuit, assuming point A to be at zero potential, use Kirchhoff's rules to determine the potential at point B and value of R .



5 MARK QUESTIONS

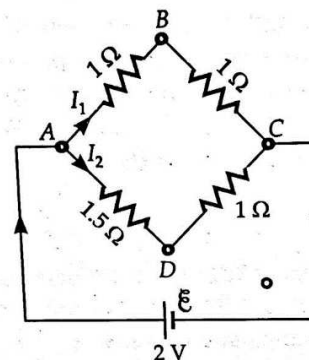
Q51. (a) The electron drift arises due to the force experienced by electrons in the electric field inside the conductor. But force should cause acceleration. Why then do the electrons acquire a steady average drift speed?

(b) If the electron drift speed is so small, and the electron's charge is small, how can we still obtain large amounts of current in a conductor?

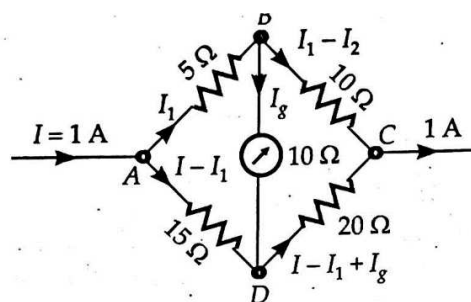
(c) When electrons drift in a metal from lower to higher potential, does it mean that all the 'free' electrons of the metal are moving in the same direction?

(d) Are the paths of electrons straight lines between successive collisions (with the positive ions of the metal) in the (i) absence of electric field, (ii) presence of electric field?

Q52. Two cells of EMF 3V and 4V and internal resistances 1 ohm and 2 ohms respectively are connected in parallel so as to send current in same direction through an external resistance of 5 ohm. Draw the circuit diagram using Kirchhoff's laws. Calculate the current through each branch of the circuit and potential difference across 5-ohm resistance.



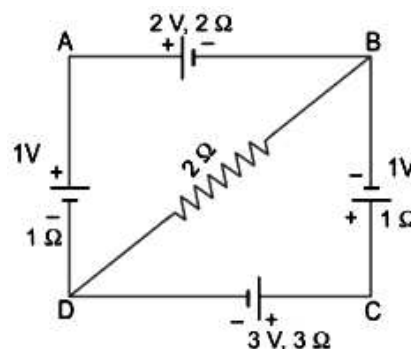
Q53. Calculate the potential difference between the junctions B and D in the Wheatstone Bridge shown in figure



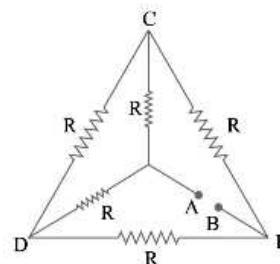
Q54. Determine the current flowing through the galvanometer G of Wheatstone Bridge shown in figure.

Q55. For the circuit given below, find the potential difference b/w points B and D.

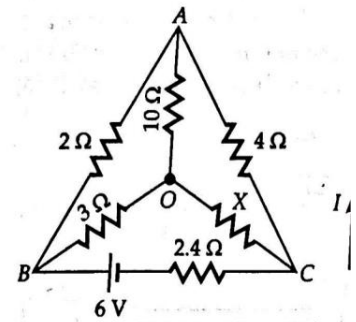
Q56. Calculate Equivalent Resistance of the given electrical network b/w points A and B. Also calculate the current through CD & ACB if a 10V dc source is connected b/w points A and B and the value of $R = 2\Omega$. If battery of V volt is connected find current in circuit and current through DC.



Q57. P, Q, R and S are four resistance wires of 2, 2, 2 and 3 ohms respectively. Find out the resistance with which S must be shunted in order that bridge may be balanced.



Q58. Find the value of unknown resistance X , in the following circuit, if no current flows through the section AO . Also calculate the current drawn by the circuit from the battery of emf $6V$ and negligible internal resistance.



SECTION-C-Answer Key

Multiple Choice Questions -solutions

1.(b)

2. (a)

3. (a)

4. Rate of heat developed, $P = V^2/R$

For given V , $P \propto 1/R = A/\rho l = \pi r^2/\rho l$

Now, $P_1/P_2 = (r_1^2/r_2^2)(l_2/l_1)$

As per question, $l_2 = l_1/2$ and $r_2 = 2r_1$

$P_1/P_2 = (1/4) \times (1/2) = 1/8$

$P_2 = 8P_1$ Answer: (a) Increased 8 times

5. Resistance after temperature increases by 500°C ,

$$R_T = \text{Voltage applied/Current} = 220/2 = 110$$

$$\text{Also, } R_T = R_0 (1 + \alpha \Delta T)$$

$$110 = 100 (1 + (\alpha \times 500))$$

$$\alpha = 10/(100 \times 500) = 2 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}$$

Answer: (b) $2 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}$

6. Resistance of a wire of length l and radius r is given by

$$R = \rho l/A = (\rho l/A) \times (A/A)$$

$$R = (\rho V/A^2) = (\rho V/\pi^2 r^4) (\because V = Al)$$

$$\text{i.e., } R \propto 1/r^4$$

$$R_1/R_2 = (r_2/r_1)^4$$

$$\text{Here, } R_1 = 100 \text{ } \Omega, r_1 = r, r_2 = r/2$$

$$R_2 = R_1(r_1/r_2)^4 = 16R_1 = 1600 \text{ } \Omega$$

Answer: (d) $1600 \text{ } \Omega$

7. Power of 15 bulbs of $40 \text{ W} = 15 \times 40 = 600 \text{ W}$

Power of 5 bulbs of $100 \text{ W} = 5 \times 100 = 500 \text{ W}$

Power of 5 fan of $80 \text{ W} = 5 \times 80 = 400 \text{ W}$

Power of 1 heater of $1 \text{ kW} = 1000$

Total power, $P = 600 + 500 + 400 + 1000 = 2500 \text{ W}$

When these combination of bulbs, fans and heater are connected to 220 V mains, current in the main fuse of building is given by

$$I = P/V = 2500/220 = 11.36 \text{ A} \approx 12 \text{ A}$$

Answer: (d) 12 A

8. (b) increase by 0.2%

$$\text{Resistance of wire } R = \rho l/A \dots\dots\dots(1)$$

On stretching, volume (V) remains constant.

$$\text{So } V = Al \text{ or } A = V/l$$

$$\text{Therefore, } R = \rho l^2/V \text{ (Using (1))}$$

Taking logarithm on both sides and differentiating we get,

$$\Delta R/R = 2\Delta l/l \text{ (Since } V \text{ and } \rho \text{ are constants)}$$

$$(\Delta R/R)\% = (2\Delta l/l)\%$$

Hence, when the wire is stretched by 0.1% its resistance will increase by 0.2%

Answer: (b) increase by 0.2%

9.(c) $P_2 > P_1 > P_3$

$$\text{For } R_1 = 1 \text{ ohm}$$

$$\text{For } R_2 = 0.5\Omega$$

$$R_3 = 2\Omega$$

As $P = V^2 / R$ and $V=3$ for each so $P \propto 1/R$

Thus, $P_2 > P_1 > P_3$.

10. (a) $I_R = I_G$

The reading of Galvanometer G is the same with switch S open or closed. This implies that closing or opening the switch does not affect the circuit. It means that no current flows through switch S. Hence, the given arrangement of resistors is a Balanced Wheatstone bridge. Hence we can remove switch S.

R and G will be in series. Hence current flows through resistor R is equal to the current flows through the galvanometer G. $\Rightarrow I_R = I_G$. Hence option (a) is correct.

11. (b)

$$I = 6-3/3 = 3/3 = 1 \text{ A}$$

$$V_A - 6 + 1 - V_B = 0$$

$$\text{therefore } V_A - V_B = 5$$

12 (c) 3E

$$I = 10 E / 10 r = E / r$$

Current through three cells $I \times 3r = 3 E$

$$13. (b) R = R_0(1 - BT)$$

14. (a) number of charge carriers can change with temperature T .

(b) time interval between two successive collisions can depend on T

15. (a) The equivalent emf ϵ_{eq} of the two cells is between ϵ_1 and ϵ_2 , i.e., $\epsilon_1 < \epsilon_{eq} < \epsilon_2$

$$16. (a) 2V, 0.5 \Omega$$

$$17. c) -5A$$

$$18. (d) (r_1/2) - r_2$$

Solution

$$I = \frac{3E}{R + r_1 + r_2}$$

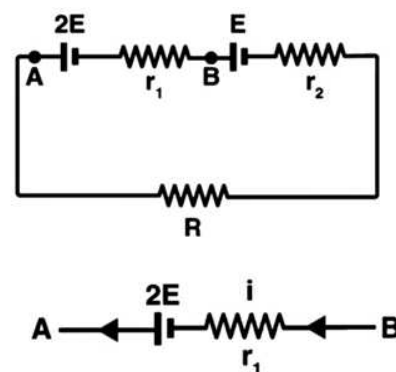
If potential difference across terminals of first cell is zero

$$V_A = V_B$$

$$2E - i r_1 = 2E - \frac{3E}{R + r_1 + r_2} r_1$$

$$2R + 2r_1 + 2r_2 = 3r_1$$

$$R = (r_1/2) - r_2$$



V.S.A

19. When area of the anode is decreased, the internal resistance of the cell increases. Lesser area of the anode decreases its tendency to attract oppositely charged ions

20. High resistivity and low temperature Coefficient of resistance

$$21. ML^2A^{-1}T^{-3}$$

22. In an alloy like nichrome (made of nickel and chromium), the free electron finds a disordered arrangement of nickel ions and chromium ions. Due to it, the electron is scattered by them randomly and very frequently. As a result of it, the value of relaxation time of electron decreases and hence resistivity increases because $\rho \propto 1/\tau$.

$$23. (c) J = \sigma E$$

The current density at any point in ohmic resistor is in the direction of electric field at that point in space having non-uniform electric field. Charges released from rest may not move along electric line of force hence statement 1 is true while statement 2 is false

24. (d) $R = R_0(1 + \alpha \Delta t)$ is valid only when the change in temperature is small and $\Delta R \ll R_0$

Statement 1 is false and 2 is true

25. (c) statement 1 is true 2 is false

26. (c) statement 1 is true 2 is false

27. (c) statement 1 is true 2 is false

SOL Internal resistance is not characteristic property but depends upon the quantity and concentration of electrolyte and distance between the electrodes

28. (a) Both true 2 is explanation of 1

2 MARKS QUESTIONS SOLUTION

29. Ans. $1.6 \times 10^6 \text{ A m}^2$; 1.25×10^4

$$j = I/A = 1.6 \times 10^6 \text{ Am}^{-2}$$

$$v_d = j/ne = 1.25 \times 10^4 \text{ ms}^{-1}$$

30.a. $A = \frac{\pi D^2}{4} = 4.524 \times 10^{-6} \text{ m}^2$

$$j = I/A = 2.21 \times 10^6 \text{ Am}^{-2}$$

$$\text{b. } A = \frac{\pi D^2}{4} = 2.01 \times 10^{-6} \text{ m}^2$$

$$v_d = I/neA = 3.7 \times 10^4 \text{ ms}^{-1}$$

31. Ans. 1495.2 kg

$$R = \frac{\rho l}{A}$$

$$\text{From which } A = 1.68 \times 10^{-5} \text{ m}^2$$

$$m = \text{volume} \times \text{density} = A \times l \times \text{density} = 1495.2 \text{ kg}$$

32. Ans: Nichrome Wire

$$P = I^2 R$$

$P \propto R$ Heat produced is higher in Nichrome wire.

33. Solution $R_0 = 5 \Omega$, $R_{100} = 5.23 \Omega$ and $R_t = 5.795 \Omega$

$$t = \frac{R_t - R_0}{R_{100} - R_0} \times 100, \quad R_t = R_0 (1 + \alpha t)$$

$$= \frac{5.795 - 5}{5.23 - 5} \times 100$$

$$= \frac{0.795}{0.23} \times 100 = 345.65^\circ \text{C}$$

34. Ans. 7.875Ω

$$n = 100$$

$$D = 0.03 \text{ m}$$

$$l = n\pi D = 100 \times \pi \times 0.03 \text{ m}$$

$$A = \pi r^2$$

$$R = \frac{\rho l}{A} = 7.875 \Omega$$

35. $v_d = (I/neA)$ Now, $e = 1.6 \times 10^{-19} \text{ C}$, $A = 1.0 \times 10^{-7} \text{ m}^2$, $I = 1.5 \text{ A}$. The density of conduction electrons, n is equal to the number of atoms per cubic metre (assuming one conduction electron per Cu atom as is reasonable from its valence electron count of one). A

cubic metre of copper has a mass of $9.0 \times 10^3 \text{ kg}$. Since 6.0×10^{23} copper atoms have a mass of 63.5 g ,

which gives

$$v_d = \frac{1.5}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 1.0 \times 10^{-7}}$$

$$= 1.1 \times 10^{-3} \text{ m s}^{-1} = 1.1 \text{ mm s}^{-1}$$

$$n = \frac{6.0 \times 10^{23}}{63.5} \times 9.0 \times 10^6$$

$$= 8.5 \times 10^{28} \text{ m}^{-3}$$

(b) (i) At a temperature T , the thermal speed* of a copper atom of mass M is obtained from $\left[\frac{1}{2} m \overline{v^2} \right] = \frac{3}{2} kT$ and is thus typically of the order of $\sqrt{\frac{3kT}{M}}$, where k is the Boltzmann constant. For copper at 300 K , this is about $2 \times 10^2 \text{ m/s}$. This figure indicates the random vibrational speeds of copper atoms in a conductor. Note that the drift speed of electrons is much smaller, about 10^{-5} times the typical thermal speed at ordinary temperatures. (ii) An electric field travelling along the conductor has a speed of an electromagnetic wave, namely equal to $3.0 \times 10^8 \text{ m s}^{-1}$

The drift speed is, in comparison, extremely small; smaller by a factor of 10^{-11}

36. Ans. $V_A - V_B = -8 \text{ volt}$.

37. Using KVL, ANS = -10 V.

38.

$$\begin{aligned}V &= E + ir \\&= 2 + 0.15 \\&= 2.15\text{V}\end{aligned}$$

39.

2 / 5 Ampere

40. On applying Kirchhoff's current law on junction A, at junction A $2 + 3 = I + 4$ so,

$$I = +1\text{A}$$

41. In the circuit when an electron approaches a junction, in addition to the uniform E that faces it normally (which keep the drift velocity fixed), as drift velocity (v_d) is directly proportional to Electric field (E). That's why there is accumulation of charges on the surface of wires at the junction.

These produce additional electric fields. These fields alter the direction of momentum. Thus, the motion of a charge across junction is not momentum conserving.

3 MARKS QUESTIONS SOLUTION

42. $v_d = (I/neA)$ Now, $e = 1.6 \times 10^{-19}\text{C}$, $A = 1.0 \times 10^{-7}\text{m}^2$, $I = 1.5\text{A}$.

The density of conduction electrons, n is equal to the number of atoms per cubic metre (assuming one conduction electron per Cu atom as is reasonable from its valence electron count of one). A cubic metre of copper has a mass of $9.0 \times 10^3\text{kg}$. Since 6.0×10^{23} copper atoms have a mass of 63.5g,

which gives

$$\begin{aligned}v_d &= \frac{1.5}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 1.0 \times 10^{-7}} \\&= 1.1 \times 10^{-3}\text{ m s}^{-1} = 1.1\text{ mm s}^{-1}\end{aligned}$$

$$\begin{aligned}n &= \frac{6.0 \times 10^{23}}{63.5} \times 9.0 \times 10^6 \\&= 8.5 \times 10^{28}\text{ m}^{-3}\end{aligned}$$

(b) (i) At a temperature T , the thermal speed* of a copper atom of mass M is obtained from $\left[\frac{1}{2}mv^2 \right] = \frac{3}{2}kT$ and is thus typically of the order of $\sqrt{kT/M}$, where k is the Boltzmann constant. For copper at 300 K, this is about $2 \times 10^2\text{ m/s}$. This figure indicates the random vibrational speeds of copper atoms in a conductor. Note that the drift speed of electrons is much smaller, about 10^{-5} times the typical thermal speed at ordinary temperatures. (ii) An electric field travelling along the conductor has a speed of an electromagnetic wave, namely equal to $3.0 \times 10^8\text{ m s}^{-1}$

The drift speed is, in comparison, extremely small; smaller by a factor of 10^{-11}

43.Length of aluminium = l_1

Resistance of aluminium = R

Resistivity of aluminium, $\rho_{Al} = \rho_1 = 2.63 \times 10^{-8} \Omega m$

Relative density of aluminium, $d_1 = 2.7$

Area of cross-section of the aluminium wire = A_1

Length of copper = l_2

Resistance of copper = R_2

Resistivity of copper, $\rho_{Cu} = \rho_2 = 1.72 \times 10^{-8} \Omega m$

Relative density of copper, $d_2 = 8.9$

Area of cross-section of the copper wire = A_2

Therefore,

$$R_1 = \rho_1 l_1 / A_1 \quad \text{and} \quad R_2 = \rho_2 l_2 / A_2$$

$$\rho_1 l_1 / A_1 = \rho_2 l_2 / A_2$$

$$\text{Given } l_1 = l_2 \quad A_1 / A_2 = \rho_2 / \rho_1 = 1.52$$

Mass of aluminium, $m_1 = \text{Volume} \times \text{density} = A_1 l_1 \times d_1$

Mass of copper = $m_2 = \text{Volume} \times \text{density} = A_2 l_2 \times d_2$

$$m_1 / m_2 = (A_1 d_1 / A_2 d_2)$$

$$m_1 / m_2 = (1.52) \times (2.7 / 8.9) = (1.52) \times (0.303) \quad , \quad m_1 / m_2 = 0.46$$

44. When the current through the element is very small, heating effects can be ignored and the temperature T_1 of the element is the same as room temperature. When the toaster is connected to the supply, its initial current will be slightly higher than its steady value of 2.68 A. But due to heating effect of the current, the temperature will rise. This will cause an increase in resistance and a slight decrease in current. In a few seconds, a steady state will be reached when temperature will rise no further, and both the resistance of the element and the current drawn will achieve steady values. The resistance R_2 at the steady temperature T_2 is

$$R_2 = 230 \text{ V} / 2.68 \text{ A} = 85.8 \Omega$$

Using the relation $R_2 = R_1 [1 + \alpha (T_2 - T_1)]$ with $\alpha = 1.70 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$,

we get $T_2 - T_1 = 820 \text{ }^\circ\text{C}$

that is, $T_2 = (820 + 27.0) \text{ }^\circ\text{C} = 847 \text{ }^\circ\text{C}$

Thus, the steady temperature of the heating element (when heating effect due to the current equals heat loss to the surroundings) is $847 \text{ }^\circ\text{C}$.

45. Ans. (a) $[I T^{-1}]$, $[I]$ and $[I T]$

(b) 23 A

Here, $q = at^2 + bt + c$

(a) Dimensions of each factor on R.H.S. of the above relation are that of charge i.e. current $\times$ time or $[IT]$.

$$[a] = \frac{q}{t^2} = [IT^{-1}]$$

$$[b] = \frac{q}{t} = [I]$$

$$[c] = [q] = [IT]$$

(b) Here, $a = 3$, $b = 5$ and $c = 2$

Now, $I = dq/dt = 2at + b$

$$I \text{ (at } t = 3 \text{ s)} = 2 \times 3 \times 3 + 5 = 23 \text{ A}$$

46. From Kirchhoff's I law $I_3 = I_1 + I_2 \dots (i)$

Applying Kirchhoff's II law to loop PRSP

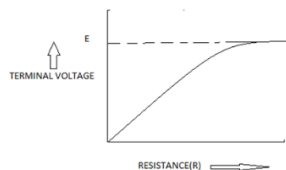
$$-20I_3 - 200I_2 + 5 = 0$$

$$\Rightarrow 40I_2 + 4I_3 = \dots (ii)$$

Applying Kirchhoff's II law to loop PRQP

$$-20I_3 - 60I_2 + 4 = 0$$

$$\Rightarrow 15I_1 + 5I_3 = 1$$



$$47. V = IR = \frac{E}{r/R + 1}$$

cell.

$V = \text{EMF}$, when no current is drawn from the

48. Ans. I) SERIES

$$E_{eq} = E_1 + E_2 + E_3 + \dots + E_n = nE$$

$$r_{eq} = r_1 + r_2 + r_3 + \dots + r_n = nr$$

$$E_{net} = 4 \times 1.4 = 5.6 \text{ V}$$

$$r_{eq} = 4 \times 0.8 = 3.2\text{-ohm}$$

$$R = 10\text{-ohm}$$

$$R_{net} = 13.2 \text{ ohm}$$

$$I = 5.6 / 13.2 = I_{series} = 0.424 \text{ A}$$

FOR PARALLEL

$$E_{eq} = E$$

$$(1/r_{eq}) = (1/r_1) + (1/r_2) + (1/r_3) \dots\dots\dots(1/r_n)$$

$$E_{parallel} = 1.4 \text{ V}$$

$$r_{eq} = .8 / 4 = .2 \text{ ohm}$$

$$R = 10 \text{ ohm}$$

$$R_{net} = 10.2 \text{ ohm}$$

$$I = 1.4 / 10.2 = I_{parallel} = 0.137 \text{ A}$$

current through each cell is 0.03 A

$$49. \text{ EMF} = E \quad R_{net} = r + 2$$

$$I = E / r + 2 = 0.5$$

$$0.5 \times 2 + 0.5 r = E \quad \dots\dots\dots 1)$$

Similarly

$$0.25 \times 5 + 0.25 r = E \quad \dots\dots\dots 2)$$

Solving

$$r = 1 \text{ ohm}, E = 1.5 \text{ V}$$

when short circuited, current = 1.5

50. In 1st loop

$$-R + 1 + 2 = 0$$

$$\Rightarrow R = 3\Omega$$

In 2nd loop

$$-2R_1 + 2 - 2 = 0 \quad \Rightarrow R_1 = 0$$

Now going from point, A to B via C.

$$V_A + 1 - 0 = V_B \quad \Rightarrow V_B = 1 \text{ V}$$

5 MARKS QUESTIONS SOLUTION

51.(a) Each 'free' electron does accelerate, increasing its drift speed until it collides with a positive ion of the metal. It loses its drift speed after collision but starts to accelerate and increases its drift speed again only to suffer a collision again and so on. On the average, therefore, electrons acquire only a drift speed.

(b) Simple, because the electron number density is enormous, $\sim 10^{29} \text{ m}^{-3}$.

(c) By no means. The drift velocity is superposed over the large random velocities of electrons.

(d) In the absence of electric field, the paths are straight lines; in the presence of electric field, the paths are, in general, curved

52.(a) $1/17$ A, $9/17$ A, $8/17$ A

(b) 2.35 V

53. In loop ABCEA

$$I_1 + I_1 = 2$$

$$I_1 = 1 \text{ A}$$

From loop ADCEA

$$1.5I_2 + I_2 = 2$$

$$I_2 = 0.8 \text{ A}$$

$$V_A - V_B = 1 \times 1 = 1 \text{ V}$$

$$V_A - V_D = 0.8 \times 1.5 = 1.2 \text{ V}$$

$$V_B - V_D = 1.2 - 1.0 = 0.2 \text{ V}$$

54. From loop ABDA

$$5I_1 + 10I_g - (I - I_1)15 = 0$$

Put $I = 1 \text{ A}$

$$4I_1 + 2I_g = 3$$

For Loop BCDB

$$10(I_1 - I_g) - 20(I - I_1 + I_g) - 10I_g = 0$$

Put $I = 1 \text{ A}$

$$3I_1 - 4I_g = 2$$

On solving

$$I_g = 1/22 = 0.0454 \text{ A}$$

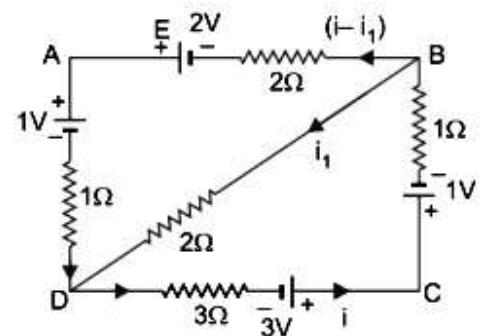
55. Applying Kirchhoff's second law to mesh BADB,

$$-2(i - i_1) + 2 - 1 - 1 \cdot (i - i_1) + 2i_1 = 0$$

$$\Rightarrow 3i - 5i_1 = 1 \dots (i)$$

Applying Kirchhoff's law to mesh DCBD,

$$-3i + 3 - 1 - 1 \times i - 2i_1 = 0$$



$$\Rightarrow 4i + 2i_1 = 2$$

$$\text{Or } 2i + i_1 = 1 \dots(\text{ii})$$

Multiplying equation (ii) with 5, we get

$$10i + 5i_1 = 5 \dots(\text{iii})$$

Adding (i) and (iii), we get

$$13i = 6 \Rightarrow i = 6/13 \text{ A}$$

$$\text{From (ii), } i_1 = 1 - 2i = 1 - 12/13 = 1/13 \text{ A}$$

Potential difference between B and D is

$$V_B - V_D = i_1 \times 2 = 2/13 \text{ V}$$

$$56. R_{AB} = 2 \text{ Ohm}$$

$$I_{CD} = 0$$

$$I_{ACB} = V/2R = 10/4 = 2.5 \text{ A}$$

$$57. \text{ Using } \frac{P}{Q} = \frac{R}{X}$$

$$X = \frac{SU}{S+U}$$

Put $X=2$, $S=3$, Find U = unknown

$$U = 3 \text{ ohm}$$

$$I = V/3$$

58. No current flows through section AO, So Wheatstone Bridge is balanced

$$2/4 = 3/X$$

$$X = 6 \text{ ohm}$$

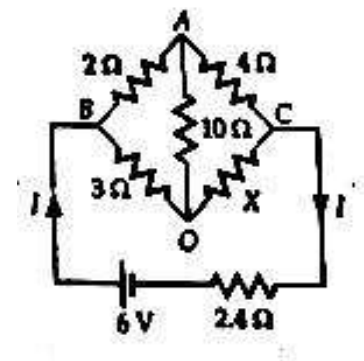
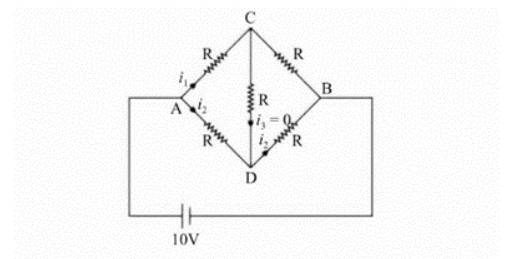
$$R_{BAC} = 6 \text{ Ohm}$$

$$R_{BOC} = 9 \text{ ohm}$$

$$R = \frac{6 \times 9}{6+9} = 3.6$$

$$\text{Total resistance} = 3.6 + 2.4 = 6$$

$$I = 6/6 = 1 \text{ A}$$



Case Based/Passage based integrated questions

59. When a conductor does not have a current through it, its conduction electrons move randomly, with no net motion in any direction. When the conductor does have a current through it, these electrons actually still move randomly, but now they tend to drift with a drift speed V_d in the direction opposite to the applied electric field that causes current. The drift speed is very small as compared to the speeds in the random motion. For example, in the copper conductors of household wiring, electron drift speeds are perhaps 10^5 m/s to 10^3 m/s, where as the random speed is around 10 ms⁻¹.

- (i) The electron drift speed is estimated to be only a few mm s⁻¹ for currents in the range of a few amperes? How is current established almost the instant a circuit is closed?
- (ii) The electron drift arises due to the force experienced by electrons in the electric field inside the conductor. But force should cause acceleration. Why do the electrons acquire a steady average drift speed?
- (iii) If the electron drift speed is so small, and the electron's charge is small, how can we still obtain large amounts of current in a conductor?
- (iv) When electrons drift in a metal from lower to higher potential, does it mean that all the 'free' electrons of the metal are moving in the same direction?
- (v) Are the paths of electrons straight lines between successive collisions (with the positive ions of the metal) in the (a) absence of electric field, (b) presence of electric field?

Case Based/Passage based integrated questions-answers

59. (i) As soon as a circuit is closed, everywhere in conductor, electric field is set up (with the speed of light), and the conduction of electron at every point experience a drift.
- (ii) Each conduction electron does accelerate, and gain speed until it collides with a positive ion of a conductor, thereby losing its drift speed after collision again it gains kinetic energy but suffers a collision again and so on. Therefore, on the average, electron acquire only a drift speed.
- (iii) As number of the density of electrons ($\approx 10^{29}$ m⁻³) is very large, therefore current flowing is large.

(iv) No. When electric field is applied, the net drift of the electrons is from lower to higher potential. But locally electrons collide with ions and may change its direction during the course of their motion.

(v) Yes. In the (a) absence of electric field, the paths are straight lines.

Reason: As electrons were not acted upon by any kind of forces.

(b) No. In the presence of electric field paths were curved. Reason: As direction of random velocities and acceleration are not always same.

Case-Based MCQ

60. Read the following text and answer the following questions on the basis of the same:

Electric Toaster: Small Industries Service Institute Takyelpat Industrial Estate Imphal has designed an Electric toaster which is operated at 220 volts A.C., single phase and available in four different rated capacity such as 600 W, 750 W, 1000 W and 1250 W. The heating element is made of nichrome 80/20 (80% nickel, 20% chromium), since Nichrome does not get oxidize readily at high temperature and have higher resistivity, so it produces more heat.

The element is wound separately on Mica sheets and fitted with body of toaster with the help of ceramic terminals.

Q. 1. Heating element of the toaster is made of:

- (A) copper (B) nichrome (C) chromium (D) nickel

Q. 2. What is meant by 80/20 Nichrome?

- (A) 80% Chromium and 20% Nickel
(B) 80% Nickel and 20% Chromium
(C) Purity 80%, Impurity 20%
(D) It is a mixture of Chromium and Nickel

Q.3. Which one will consume more electricity?

- (A) 600 W (B) 1000 W (C) 750 W (D) 1200 W

Q.4. Operating voltage of the device is: (A) 220 V AC, single phase

- (A) 220 V AC, single phase (B) 220 V AC, three phase (C) 220 V DC (D) 220 V AC/DC

Q. 5. Insulating materials used in the device are:

- (A) Mica (B) Ceramic (C) Mica, ceramic, Nichrome (D) Mica, ceramic

Case Based/Passage based integrated questions-answers

60.

1.Ans. Option (B) is correct.

Explanation: The heating element is made of nichrome 80/20 (80% nickel, 20% chromium).

2.Ans. Option (B) is correct.

Explanation: Nichrome 80/20 means an alloy of 80% nickel, 20% chromium.

3.Ans. Option (D) is correct.

Explanation: Electricity consumption is measured by kWh. So, 1200W toaster will consume more electricity

4.Ans. Option (A) is correct.

Explanation: The designed electric toaster is IN operated at 220 volts A.C., single phase.

5.Ans. Option (D) is correct.

Explanation: The element is wound separately on Mica sheets and fitted with body of toaster with the help of ceramic terminals.

Case-Based MCQ

61. Whenever an electric current is passed through a conductor, it becomes hot after some time. The phenomenon of the production of heat in a resistor by the flow of an electric current through it is called heating effect of current or Joule heating. Thus, the electrical energy supplied by the source of emf is converted into heat. In purely resistive circuit, the energy expended by the source entirely appears as heat. But if the circuit has an active element like a motor, then a part of energy supplied by the source goes to do useful work and the rest appears as heat. Joule's law of heating forms the basis of various electrical appliances such as electric bulb, electric furnace, electric press etc.

(i) Which of the following is correct statement?

- (a) Heat produced in a conductor is independent of the current flowing.
- (b) Heat produced in a conductor varies inversely as the current flowing.
- (c) Heat produced in a conductor varies directly as the square of the current flowing.
- (d) Heat produced in a conductor varies inversely as the square of the current flowing.

- (ii) If the coil of a heater is cut to half, what would happen to heat produced?
 (a) Doubled (b) Halved (c) Remains same (d) Becomes four times.
- (iii) A 25 W and 100 W are joined in series and connected to the mains. Which bulb will glow brighter?
 (a) 100 W (b) 25 W (c) Both bulbs will glow brighter (d) None will glow brighter
- (iv) A rigid container with thermally insulated wall contains a coil of resistance $100\ \Omega$, carrying 1A. Change in its internal energy after 5 min will be
 (a) 0 KJ (b) 10 KJ (c) 20 KJ (d) 30 KJ
- (v) The heat emitted by a bulb of 100 W in 1 min is
 (a) 100 J (b) 1000 J (c) 600 J (d) 6000 J

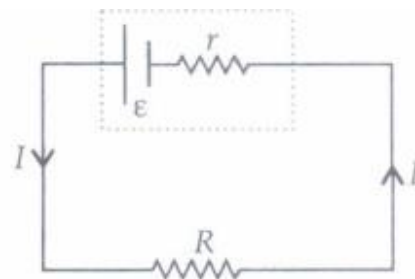
Case Based/Passage based integrated questions-answers

61. (i) (c) Heat produced in a conductor varies directly as the square of the current flowing
 (ii) (a) Doubled
 (iii) (a) 100 W
 (iv) (d) 30 kJ
 (v) (d) 6000 J

Case Based/Passage based integrated questions

62. Emf of a cell is the maximum potential difference between two electrodes of the cell when no current is drawn from the cell. Internal resistance is the resistance offered by the electrolyte of a cell when the electric current flows through it. The internal resistance of a cell depends upon the following factors;

- (i) distance between the electrodes
 (ii) nature and temperature of the electrolyte
 (iii) nature of electrodes
 (iv) area of electrodes.



For a freshly prepared cell, the value of internal resistance is generally low and goes on increasing as the cell is put to more and more use. The potential difference between the two electrodes of a cell in a closed circuit is called terminal potential

difference and its value is always less than the emf of the cell in a closed circuit. It can be written as $V = E - Ir$.

(i) The terminal potential difference of two electrodes of a cell is equal to emf of the cell when

- a) during charging b) during discharging c) both a and b d) $I = 0$

(ii) A cell of emf ϵ and internal resistance r gives a current of 0.5 A with an external resistance of 12Ω and a current of 0.25 A with an external resistance of 25Ω . internal resistance of the cell and emf of the cell are

- a) 1Ω , 6.5 V b) 5Ω , 6.5 V c) 1Ω , 7.5 V d) 5Ω , 7.5 V

(iii) Choose the wrong statement.

(a) Potential difference across the terminals of a cell in a closed circuit is always less than its emf.

(b) Internal resistance of a cell decreases with the decrease in temperature of the electrolyte.

(c) Potential difference versus current graph for a cell is a straight line with a -ve slope

(d) Terminal potential difference of the cell when it is being charged is given as $V = E + Ir$.

(iv) An external resistance R is connected to a cell of internal resistance r , the maximum current flows in the external resistance, when

- (a) $R = r$ (b) $R < r$ (c) $R > r$ (d) $R = 0$

(v) IF external resistance connected to a cell has been increased to 5 times, the potential difference across the terminals of the cell increases from 10 V to 30 V. Then, the emf of the cell is

- (a) 30 V (b) 60V (c) 50 V (d) 40 V

SOLUTIONS:

62 (i) d) $I = 0$

(ii) a) 1Ω , 6.5 V

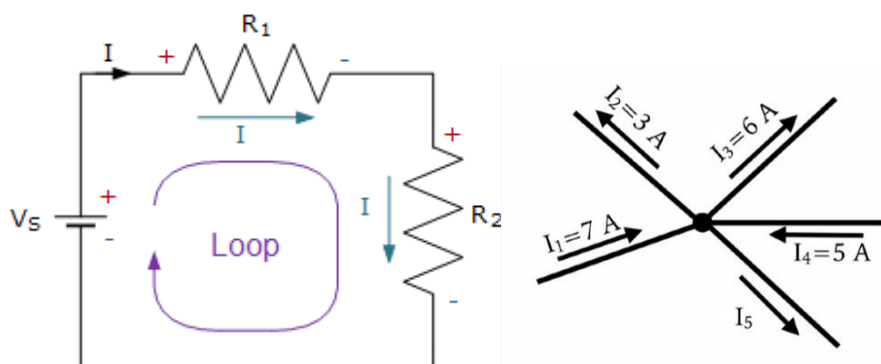
(iii) (b) Internal resistance of a cell decreases with the decrease in temperature of the electrolyte.

(iv) (d) $R = 0$

(v) (b) 60V

63. KIRCHHOFF'S RULES

Electric circuits generally consist of a number of resistors and cells interconnected sometimes in a complicated way. The formulae we have derived earlier for series and parallel combinations of resistors are not always sufficient to determine all the currents and potential differences in the circuit. Two rules, called Kirchhoff's rules, are very useful for analysis of electric circuits. Given a circuit, we start by labelling currents in each resistor by a symbol, say I , and a directed arrow to indicate that a current I flows along the resistor in the direction indicated. If ultimately, I is determined to be positive, the actual current in the resistor is in the direction of the arrow. If I turns out to be negative, the current actually flows in a direction opposite to the arrow. Similarly, for each source (i.e., cell or some other source of electrical power) the positive and negative electrodes are labelled, as well as, a directed arrow with a symbol for the current flowing through the cell.



The directed sum of the potential differences (voltages) around any closed loop is zero.

1. Kirchhoff 1ST Law is conservation of

- a) Charge b) Energy c) Potential d) Momentum

2. Kirchhoff junction rule can be written as

- a) $\sum V = 0$ b) $\sum I = 0$ c) $\sum R = 0$ d) $\sum q = 0$

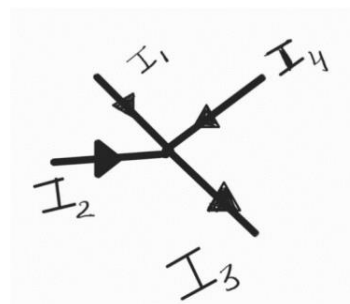
3. Kirchhoff's Voltage Law is the conservation of

- a) Energy b) Charge c) Current d) Momentum

4. Kirchhoff's Voltage Law is applied over

- a) Closed Circuit loop b) At a circuit node
c) Across battery d) None of the above

5. Find the current I_3 when $I_1=2A$, $I_2=9A$, $I_4=4A$

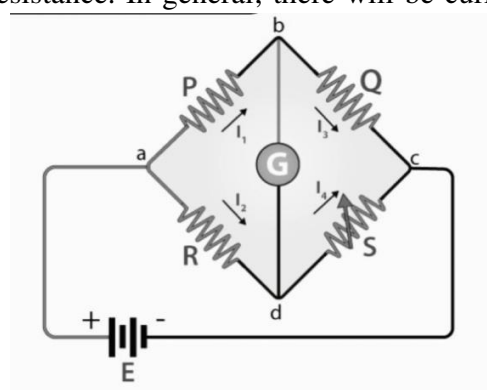


SOLUTIONS:

1. a) Charge
2. b) $\Sigma I = 0$
3. a) Energy
4. a) Closed Circuit loop
5. 15A

Case Based/Passage based integrated questions

Q64. As an application of Kirchhoff's rules consider the circuit shown, which is called the Wheatstone bridge. The bridge has four resistors R_1 , R_2 , R_3 and R_4 . Across one pair of diagonally opposite points (A and C in the figure) a source is connected. This (i.e., AC) is called the battery arm. Between the other two vertices, B and D, a galvanometer G (which is a device to detect currents) is connected. This line, shown as BD in the figure, is called the galvanometer arm. For simplicity, we assume that the cell has no internal resistance. In general, there will be currents flowing across all the resistors as well as a current I_g through G. Of special interest, is the case of a balanced bridge where the resistors are such that $I_g = 0$. We can easily get the balance condition, such that there is no current through G.



1. What is the principle of the Wheatstone bridge?
2. Name the instrument that is used as a null detector in the Wheatstone bridge.
3. Which among the following is a false statement?
a) A galvanometer is used as the null detector in a Wheatstone bridge

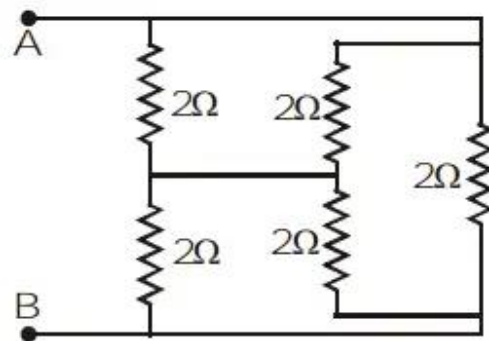
- b) A galvanometer is an ammeter with low resistance in series
 c) Wheatstone bridge is susceptible to high dc current
 d) Due to the errors introduced in contact resistance, a Wheatstone bridge cannot be used for accurate measurement

4. $PR = QS$ is the equation of a balanced Wheatstone bridge. Is it true or false?

- a) True b) False

5. The equivalent resistance across AB is :

- a) $1\ \Omega$
 b) $2\ \Omega$
 c) $3\ \Omega$
 d) $4\ \Omega$



SOLUTIONS:

3. c) Wheatstone bridge is susceptible to high dc current
 4. b) False
 5. a) $1\ \Omega$

Here CDEF will represent a balanced Wheatstone bridge. As the each arm resistance is equal to $2\ \Omega$ so equivalent resistance for bridge is $R_{eq} = 2\ \Omega$. Thus, we can remove the conducting wire GH and the corresponding circuit is as shown in figure.

Equivalent resistance between A and B is $R_{AB} = 2 || 2 = 2/2 = 1\ \Omega$

